

Momentum at Long Holding Periods*

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Abstract

This paper examines how the one-month lag embedded in the academic definition of momentum provides useful information for identifying future momentum stocks. Building on this insight, we design portfolio formation rules that sustain strong momentum exposure over long horizons. Compared with standard methodologies for momentum, our strategies (1) reduce turnover, (2) lower risk, (3) increase capacity, and (4) raise returns. Using conservative assumptions linking turnover to trading costs, these strategies increase the net (of trading costs) annual returns by more than five percentage points and enhance the resilience of momentum to post-publication return decay.

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1. Introduction

Well-functioning financial markets are characterized by their unpredictability. Any information that enables investors to anticipate future price movements, even imperfectly, is valuable. Yet, within this stochastic environment, a unique feature of the momentum anomaly allows investors to forecast which stocks will exhibit momentum in the future. In this paper, we explore how investors can leverage this insight to design trading strategies that maintain strong momentum exposure over long holding periods while substantially reducing portfolio turnover.

The predictability of future momentum arises from a feature of its definition: the exclusion of month $t-1$ from the formation rule, because stocks experience reversals at short horizons (Jegadeesh 1990). Specifically, at the end of each month $t-1$, the standard 12-2 momentum strategy sorts stocks based on their cumulative returns from month $t-12$ to month $t-2$ and takes long (short) positions in stocks in the top (bottom) decile (e.g., Jegadeesh and Titman 1993, Asness 1994).^{1,2} Therefore, we know today which stocks will be in the top and bottom deciles next month based on cumulative returns from month $t-11$ to month $t-1$. We can also reliably predict which stocks will be in the momentum portfolio beyond next month. For example, momentum rankings based on returns from month $t-10$ to month $t-1$ have a correlation of 0.94 with the momentum rankings in two months, and rankings based on returns from month $t-6$ to month $t-1$ have a correlation of 0.72 with the momentum rankings in six months.

We use this information to address two shortcomings of the standard 12-2 momentum strategy, as highlighted in Figure 1: high turnover when rebalanced monthly and rapid return decay when rebalanced less frequently. Specifically, the high returns documented in the academic literature often overlook trading costs. While a monthly-rebalanced momentum strategy—commonly used in academic studies—has produced an impressive 15.52% annual return, it also requires an extreme 939% annual turnover. It is therefore unclear to what extent a real-world

¹ We adopt the convention described on Ken French's web site and sort stocks at the end of month $t-1$ and compute returns over month t . Therefore, today refers to the end of month $t-1$ (beginning of month t). The 12-2 momentum strategy ranks stocks based on cumulative returns from month $t-12$ to month $t-2$ (inclusive).

² This decile-sorted momentum strategy is widely used in the empirical literature studying momentum trading strategies and anomalies. This approach differs from the momentum factor used in Carhart (1997), which is constructed using 30% breakpoints, and from the Fama–French UMD factor, which is based on 2×3 sorts on size and momentum. While these factors have somewhat lower turnover, the decile-sorted strategy is standard in studies examining momentum trading strategies and allows us to fully exploit the cross-sectional dispersion in momentum ranks. Importantly, our results are not driven by this specific construction, and similar patterns arise when alternative momentum definitions are considered.

investor can still capture meaningful returns from momentum once the costs of such high turnover are accounted for.³ Moreover, research by McLean and Pontiff (2016) and Calluzzo et al. (2019) shows that anomaly returns decline by over 50% after academic publication.⁴ It is perhaps not surprising that faced with these costs and uncertainty about future returns, momentum investors often choose to rebalance at slower frequencies.⁵ Prior studies have also proposed several approaches to mitigate trading costs, including reducing rebalancing frequency (e.g., Novy-Marx and Velikov 2016). For instance, we find that rebalancing semi-annually reduces annual turnover by a factor of three, to 313%. However, this slower rebalancing comes at a large cost to returns, which drop to 10.72%, nearly 5% lower compared to monthly rebalancing.

In this paper, we address the tradeoff between rebalancing frequency and returns by exploiting the predictability of future momentum to develop strategies that reduce the return decay of momentum over longer holding periods. We begin with a simple thought experiment: if we know a stock will exit the momentum portfolio next month, what happens if we exclude it from the momentum portfolio this month? We test this idea, which we term the *filter 11-1* momentum portfolio, and find that it provides two key advantages. First, because its holdings are a subset of the monthly rebalanced 12-2 momentum portfolio for the two-month period, it can be rebalanced every other month, which lowers turnover to 615% (vs. 939% for the monthly rebalanced 12-2 momentum portfolio), while still maintaining high exposure to momentum. Second, as shown in Figure 2, by maintaining higher momentum exposure for longer, the filter 11-1 portfolio outperforms 12-2 momentum when both are rebalanced every other month (15.08% vs 14.79%). This outperformance grows even larger at slower rebalancing frequencies. For example, if we rebalance every 6 months, the difference is 1.91% (12.63% vs 10.72%) and is statistically significant at the 1% level.

We further refine the filter 11-1 strategy to exclude month $t-1$ to avoid stocks that are experiencing short-term reversals (Jegadeesh 1990), at the cost of less accurately predicting next month's momentum stocks.⁶ Specifically, we filter stocks from the 12-2 momentum portfolio that

³ Lesmond et al. (2004) find no returns to momentum after costs, and Patton and Weller (2020) show that the typical mutual fund earns no returns to momentum. However, Frazzini et al. (2018) find that factor strategies are still profitable after costs using a proprietary dataset from a single institutional asset manager (AQR).

⁴ In our sample, since the publication of momentum in 1993, the return of the 12-2 strategy has fallen by 41.7%.

⁵ For example, the Invesco S&P 500 Momentum ETF rebalances its holdings every 6 months (For details, see <https://www.invesco.com/us/financial-products/etfs/product-detail?audienceType=Investor&ticker=SPMO>).

⁶ Specifically, the 11-2 portfolio has only a 93.4% correlation with next month's momentum portfolio.

are not in the 11-2 momentum portfolio, which we term the *filter 11-2* momentum portfolio. As shown in Figure 2, this approach significantly outperforms 12-2 momentum at every rebalancing horizon. For example, rebalancing monthly, it produces a 16.62% return (vs. 15.52% for 12-2 momentum), and rebalancing quarterly it produces a 15.11% return (vs. 13.30% for 12-2 momentum).

These results highlight how an investor can use simple rules to modify the academic definition of momentum, based on information about the future momentum status of stocks that are known today, and improve returns and lower trading costs. This finding motivates us to develop strategies that attempt to generalize the information about the future momentum status of stocks beyond next month. We propose two distinct momentum strategies: a concentrated approach that filters out stocks that are likely to subsequently exit the momentum portfolio, and a more diversified approach that creates a blended momentum signal based on a stock's current momentum and its anticipated future momentum. Both strategies exploit the fact that the returns used to build momentum portfolios beyond next month are partially known today, and that ranks based on these returns are highly correlated with future momentum ranks.

The concentrated strategy generalizes the filter 11-2 approach. For each rebalancing horizon k , we define the *filter k* portfolio as the set of stocks that are always in the long leg or short leg over every window from 12-2 to $(12-k+1)$ -2.⁷ So, for example, the filter 3 portfolio goes long stocks that are in the top decile of returns over the 12-2, 11-2, and 10-2 windows, and short stocks that are in the bottom decile of returns over each of those windows (and the filter 2 portfolio is identical to the filter 11-2 approach). We find that these filtered strategies deliver high returns that dominate the 12-2 momentum over every horizon. The filter 8 strategy, which is rebalanced every eight months and requires 311% annual turnover, produces the highest returns; it generates a return of 17.57%, which is more than 2% higher than the return of monthly-rebalanced 12-2 momentum. Using assumptions on the relation between portfolio turnover and trading costs that are consistent with the existing literature (e.g., Schmidt et al. 2017, Frazzini et al. 2018), we find that the filter 8 strategy produces a net return of 16.02%, which is 5.19% larger than the net return of monthly-rebalanced 12-2 momentum.⁸

⁷ Hereafter, the number after filtered or blended refers to the rebalancing frequency of the strategy.

⁸ As robustness, we consider alternative assumptions on the magnitude of the transaction costs and find similar results.

We next propose a more diversified blended approach, which forms a momentum signal by equally weighting the returns of each stock over each window before ranking the stocks. Specifically, for rebalancing horizon k , we build the *blended k* strategy by assigning $1/k$ weight to return in each window from the 12-2 window to the $(12-k+1)$ -2 window before ranking stocks into deciles. For example, an investor with a four-month rebalancing horizon ($k=4$) would assign $\frac{1}{4}$ weight to the return for each of the 12-2, 11-2, 10-2, and 9-2 windows before ranking stocks into deciles and going long the stocks in the top decile, and short the stocks in the bottom decile. We find that returns of the blended momentum strategy significantly exceed those of 12-2 momentum at horizons beyond three months. For example, at a six-month rebalancing horizon the blended 6 strategy produces a 12.78% return, compared to 10.72% for 12-2 momentum rebalanced every six months.

Figure 3 presents the growth of \$1 invested in the best performing filtered and blended strategies compared to 12-2 momentum rebalanced monthly over our sample period, highlighting the economically significant outperformance of those strategies. \$1 invested in the filter 9 strategy grew to \$84,352 (a CAGR of 12.68%). That is 135.3 times larger than the growth of \$1 invested in 12-2 momentum rebalanced monthly, which is \$623 (a CAGR of 7.01%). \$1 invested in the blended 4 strategy grew to \$3,023 (a CAGR of 8.80%), which is 4.85 times greater than the amount from investing in the 12-2 momentum strategy.

As in Jegadeesh and Titman (1993), we use portfolios with overlapping holding periods, which increases capacity and reduces risk. For example, whereas the monthly-rebalanced 12-2 momentum holds on average 345 stocks and has an annualized standard deviation of 26.30%, the blended 6 strategy holds on average 731 stocks and has an annualized standard deviation of 22.97%. These attributes may make the blended approach attractive to real-world investors, despite the fact that it produces lower returns than the filtered approach. While the filtered strategies also use overlapping portfolios, the increase in holdings at less frequent rebalancing is approximately offset by the more restrictive nature of the filter. As a result, the filtered approach average holdings range from 287 stocks (at the 11-month rebalancing frequency) to 371 stocks (at the four and five-month rebalancing frequency). Nevertheless, the filtered approach still benefits from the overlapping portfolios approach, as otherwise it would become extremely concentrated in a small number of stocks at long horizons.

We perform additional tests to better understand the mechanisms that drive the blended and filtered momentum returns, and to account for other considerations relevant for a real-world investor implementing these strategies. First, we examine if the returns of these strategies are driven by increased exposure to the momentum factor or by alpha using a Carhart (1997) 4-factor regression framework. We find that the blended and filtered approaches maintain significantly higher exposure to the momentum factor than 12-2 momentum at corresponding rebalancing frequencies, which aligns with the objective of our approach to maintain momentum exposure over long horizons. We also find that the alpha generated by the blended momentum strategies is similar to the 12-2 momentum alpha at matching rebalancing horizons, but that the filtered approach generates significantly larger alpha than both. For example, the filter 8 strategy produces an alpha of 8.20%, compared to 2.61% for the blended 8 strategy, and 1.71% for the 12-2 momentum strategy rebalanced every 8 months.

Real-world investors must also consider the uncertainty regarding the level of momentum returns going forward given the large post-publication decay documented by McLean and Pontiff (2016) and Calluzzo et al. (2019). Since the publication of momentum in 1993, the return on 12-2 momentum has dropped by 41.7% to 12.66% compared to 21.71% during the sample period of the original study (1965-1989). We then examine the returns of 12-2, blended, and filtered momentum strategies in the period after the academic publication of momentum. We find that blended and filtered momentum continue to outperform 12-2 momentum in the post-publication period. In fact, the net returns of blended and filtered strategies decay less than the net returns of 12-2 momentum. This result is consistent with the lower trading costs of these strategies eating into a smaller proportion of the returns and thus providing them with a larger buffer to weather the post-publication return decay.

We also examine the performance of the long legs of the blended and filtered portfolios separately. This is motivated by the fact that many investors are unable or unwilling to short stocks (Calluzzo et al. 2025), as is evident by the disproportionate number of long-only momentum funds, compared to long-short funds, that are marketed to retail investors. The results show that the long-only implementations of the blended and filtered portfolios also produce higher returns and Sharpe ratios⁹ than long-only 12-2 momentum at matching rebalancing horizons after accounting for potential trading costs.

⁹ Throughout the paper, Sharpe ratio is computed using returns net of estimated trading costs.

We also consider if our strategies expose investors to the momentum crash risk documented by Daniel and Moskowitz (2016). We measure the drawdown size and length of each strategy. We find that the blended and filtered approaches decrease the median drawdown size and length. For example, 12-2 momentum rebalanced monthly (every six months) experiences a median drawdown size of 44.97% (45.25%) and a median drawdown length of 79 (52.5) months. In contrast, at a six-month rebalancing horizon the blended (filter) 6 strategy experiences a median drawdown size of 29.50% (24.63%) and a median drawdown length of 30 (19) months. We also measure higher-moment risk and find that all three strategies experience similar levels of negative skewness and positive kurtosis, which is consistent with the documented crash risk exposure.

We also consider concentrated versions of the 12-2 and blended momentum strategies to determine the extent to which the outperformance of the filtered strategy is driven by concentration in high-momentum stocks vs. correlation with future momentum. Specifically, each month we rank stocks by their momentum, but instead of investing in the top and bottom deciles based on those ranks, we construct more concentrated portfolios that contain the same number of stocks in the long and short legs as in the filtered approach. This approach improves the returns and Sharpe ratios of the 12-2 and blended approaches, suggesting some of the outperformance of the filtered approach is driven by its concentration. However, we still find that the filtered approach produces a higher maximum return and Sharpe ratio than the concentrated 12-2 or blended strategies.

We also examine the returns of the 12-2, filtered, and blended momentum strategies in a sample of international stocks from 46 countries in developed and emerging markets. We find that all three strategies generate positive net returns and Sharpe ratios, and that the filtered and blended approaches produce higher returns and Sharpe ratios than the 12-2 momentum strategy at every rebalancing horizon. These results can be interpreted as an out-of-sample test and improve confidence in our methodology. Furthermore, given that trading costs are higher in international markets compared to the U.S. (e.g., Domowitz et al. 2001 and Frazzini et al. 2018), it is likely that an investor implementing momentum strategies in international stocks stands to gain more from adopting the filtered and blended momentum strategies that enable them to maintain exposure to momentum at long horizons, reducing the need to rebalance and the associated trading costs.

Lastly, we explore whether machine learning techniques can improve upon our strategies. Gu et al. (2020) show that these methods can provide significant predictive gain by allowing nonlinear predictor interactions missed by other methods, and that neural network models are

among the most effective at predicting stock returns. To mimic momentum’s formation rules, we train a neural network model that uses each stock’s past year of returns to predict returns for investment horizons from one to twelve months. We use data from the pre-publication period to train the model, and assess its performance in the post-publication period. We find that the machine learning approach outperforms 12-2 momentum, but the difference is not statistically significant. However, it does not outperform the more intuitive, tractable, and simpler filtered and blended strategies.

This paper contributes to the large literature on momentum (see Subrahmanyam 2018 and Wiest 2023 for a review of the literature). Many papers attempt to explain the momentum anomaly as driven by behavioral factors (e.g., Barberis et al. 1998, Hong and Stein 1999, Grinblatt and Han 2005, Frazzini 2006, Da et al. 2014), rational risk-based explanations (e.g., Berk et al. 1999, Johnson 2002, Hou et al. 2015), institutional factors (e.g., Lou 2012, Vayanos and Woolley 2013, Chu et al. 2022), exposure to factor momentum (e.g., Ehsani and Linnainmaa 2022, Arnott et al. 2021) or liquidity and trading-cost effects (e.g., Lesmond et al. 2004, Sadka 2006, Novy-Marx and Velikov 2016). Another strand of the literature proposes strategies that improve the traditional momentum by focusing on a particular window of past returns (e.g., Novy-Marx 2012), stocks near their 52-week high (e.g., George and Hwang 2004), stocks with price changes that occur gradually (e.g., Da et al. 2014), interactions with reversal factors (e.g., Han et al. 2016), or conditioning on high and low investor sentiment (e.g., Stambaugh et al. 2012 and Antoniou et al. 2013).

Most of the literature focuses on short-term (i.e., one month) predictability, which is difficult to implement in practice due to high turnover. We focus on predictability for up to 12 months and provide easily implementable strategies that generate higher returns, lower turnover, and greater capacity than standard momentum approaches, and they remain effective even after accounting for trading costs and post-publication return decay. We also contribute to a puzzle in the retail investment space, as to why momentum has not gained the widespread popularity of some of the other anomalies documented by the academic literature such as quality and value.¹⁰ For example, Broman and Moneta (2024) report that momentum ETFs manage a total of \$12.1

¹⁰ There are several other potential explanations for the lack of retail interest in momentum investing. For example, its ambiguous intuition may deter investors, who are primed to “buy low and sell high” rather than “buy high and sell higher,” from committing to the strategy. Furthermore, its lack of a straightforward risk-based explanation may seed doubt about its persistence out of sample.

billion, compared to \$324.7 billion in value ETFs, \$77.7 billion in quality ETFs, and \$35.7 billion in low-volatility ETFs.¹¹ Funds that can mitigate momentum return decay through less frequent rebalancing may improve returns and attract greater AUM.

2. Data

We begin our empirical analysis by replicating the standard momentum strategy introduced by Jegadeesh and Titman (1993), which sorts stocks at the end of month $t-1$ based on their cumulative returns from month $t-12$ to month $t-2$ and goes long stocks that are in the top decile and short stocks that are in the bottom decile in month t .¹² This decile-sorted implementation has become standard in the empirical momentum literature (e.g., Jegadeesh and Titman 2001, Daniel and Moskowitz 2016, Novy-Marx 2012, Da et al. 2014). Our sample includes U.S. common stocks traded on the NYSE, AMEX, and NASDAQ from January 1927 to December 2021 (December 2022 for subsequent returns). We exclude utilities, financial firms, and stocks priced under \$5. Table 1 reports the performance characteristics of this investment strategy.¹³ The academic literature has focused on the one-month window as the investment horizon. Consistent with the literature, we find that the strategy is profitable, producing 15.52% annual return.

A hurdle to achieving the momentum returns documented in the literature is the high portfolio turnover required to rebalance the long and short legs of the strategy. Specifically, each month new winners must be added to the long leg of the portfolio, and stocks that are no longer winners must be removed. This process must also be repeated for the losers in the short leg. This monthly rebalancing results in the strategy having 939% annual turnover.

¹¹ Lettau et al. (2021) find that only 4% of U.S. active equity mutual funds tilt their portfolio towards momentum. Using data from Vanda Track we compute the weekly retail trading activity as buy volume plus sell volume divided by shares outstanding. We find that retail investors prefer to trade loser stocks, which is the opposite approach of a momentum strategy. This evidence is consistent with recent studies that show that retail investors are contrarian traders (e.g., Pagano et al. 2021 and Luo et al. 2025).

¹² We use the same filters as those on Ken French's web site. Specifically, to be included in the 12-2 momentum portfolio for month t , a stock must have a price for the end of month $t-13$ and a valid return for $t-2$. In addition, any missing returns from $t-12$ to $t-3$ must be -99.0, CRSP's code for a missing price. Each included stock also must have market equity data for the end of month $t-1$. We apply these filters to all the other momentum strategies in the paper to get the final sample.

¹³ We follow Gu et al. (2020) and calculate roundtrip turnover as

$$\text{Roundtrip Turnover} = \frac{1}{T} \sum_{t=1}^T \left(\sum_i \left| w_{i,t+1} - \frac{w_{i,t}(1 + r_{i,t+1})}{1 + \sum_j w_{j,t} r_{j,t+1}} \right| \right)$$

where $w_{i,t}$ is the weight of stock i in the portfolio at time t . We divide this value by two, so that our measure is consistent with how turnover is reported in the mutual fund industry and can be interpreted as the fraction of the portfolio that is replaced during each period.

One way to reduce turnover, which has been adopted by practitioners of momentum investing (e.g., Invesco S&P 500 Momentum ETF), is to rebalance the momentum portfolio at a frequency that is slower than one month. We follow the methodology of Jegadeesh and Titman (1993) and form portfolios with overlapping holding periods. Therefore, in any given month t , the strategy holds a series of portfolios that are selected in the current month as well as in the previous $K-1$ months, where K is the holding period, and the weight assigned to each portfolio is $1/K$.¹⁴ Table 1 shows that portfolio turnover drops from 939% for monthly rebalancing to 481% for quarterly rebalancing, 313% for semi-annual rebalancing, and 195% for annual rebalancing. However, rebalancing less frequently also results in a large decay in the returns earned by the momentum strategy. Returns decline from 15.52% for monthly rebalancing to 13.30% for quarterly rebalancing, 10.72% for semi-annual rebalancing, and 5.23% for annual rebalancing.

For real-world investors, over which horizon to rebalance a momentum portfolio is a function of (at least) two factors: (1) their trading costs; and (2) the extent to which they expect momentum returns will persist out-of-sample. If we assume an investor incurs 0.25% trading costs¹⁵ for every 100% roundtrip turnover, then we find that rebalancing a portfolio every two months produces the highest return (11.73% vs. 10.83% for monthly rebalanced momentum).¹⁶

Rebalancing less frequently also produces risk and capacity benefits. This is because the momentum strategy can be implemented using an overlapping portfolios approach that increases the number of stocks held by the strategy and addresses rebalance timing luck (Hoffstein et al., 2020). For example, when we rebalance the portfolio every two months, we can split the portfolio into two halves, one half is rebalanced on the first (and third and fifth ...) month and held for two months, and the other half is rebalanced on the second (and fourth and sixth ...) month and held for two months. Therefore, rebalancing every other month holds on average 452 stocks and has a standard deviation of 25.37%, compared to rebalancing every month, which holds on average 345 stocks and has a standard deviation of 26.30%. This difference increases as we rebalance less frequently and split the strategy across additional tranches. For example, semi-annual rebalancing

¹⁴ Using overlapping portfolios increases the number of stocks held by the strategy and thus increases strategy capacity and can reduce risk. These advantages are explored further later in the paper.

¹⁵ Schmidt et al. (2017) also use a fixed-cost assumption for trading costs although it changes based on the size of the stocks. They use 30 basis points for small stocks before 2001 and 40 basis points from 2001 on. For big stocks they use 15 basis points as trading costs for the whole period.

¹⁶ For robustness, we replicate our main results assuming 0.125% trading costs for every 100% roundtrip turnover, 0.375% trading costs for every 100% roundtrip turnover, and using the effective bid-ask spread measure proposed by Hasbrouck (2009) and adopted by Novy-Marx and Velikov (2016).

holds on average 746 stocks and has a standard deviation of 22.93%. Because of the reduction of risk associated with slower rebalancing, we find that the decay in the Sharpe ratio as the rebalancing horizon increases is less pronounced than the decay in returns, although we still find that rebalancing every two months has the highest Sharpe ratio (0.462) of all the rebalancing horizons.

3. Results

3.1 $T+1$ Momentum

In this section, we consider alternative approaches that reduce the return decay associated with slower rebalancing, and that can thus improve the net (of trading costs) returns achievable by real-world investors. We introduce new formation rules that increase exposure to momentum returns under less frequent rebalancing. In other words, for a given rebalancing horizon, we create portfolios that are more highly correlated with the 12-2 momentum strategy rebalanced monthly compared to a 12-2 momentum portfolio rebalanced less frequently.

A unique feature of the standard momentum methodology allows us to predict future momentum portfolios, and construct strategies that maintain high correlations to those portfolios. Specifically, the momentum formation rule ranks stocks based on an 11-month window that starts at month $t-12$ and ends at month $t-2$. The most recent month $t-1$ is excluded because it is associated with negative returns.¹⁷ This exclusion means that at the end of month $t-1$, we know the constituents of the momentum portfolio for month $t+1$, which can be identified by sorting stocks based on their returns from $t-11$ to $t-1$.¹⁸ We explore strategies that can exploit this information about the future to improve momentum returns for real-world investors who are concerned about trading costs and the future persistence of momentum returns.

3.1.1 Filter 11-1 Momentum

We first apply a simple filter to the 12-2 momentum strategy. We consider an investor forming a portfolio at the end of month $t-1$ who is conscientious about trading costs and wants to

¹⁷ In our sample, a strategy that goes long the top decile of stocks and short the bottom decile of stocks based on month $t-1$ returns generates a -0.48% return over the following month.

¹⁸ In practice, only 96.8% (93.1%) of the top (bottom) decile by 12-2 momentum next month, is in the top (bottom) decile of 11-1 momentum this month. This figure is less than 100% because the universe of firms (and hence the decile cutoffs) change between month $t-1$ and t mostly due to data filters, which we detail in the data section of the paper.

incorporate information about future momentum predictability into their formation rules. A straightforward way to reduce trading costs is to avoid taking positions in stocks in month t that they know will exit the 12-2 momentum portfolio in month $t+1$. They can identify these stocks by examining the 11-1 momentum portfolio, which will become the 12-2 momentum portfolio next month. Panel A of Table 2 reports the returns and turnover of this strategy, which we term the *filter 11-1* strategy, at different rebalancing horizons. The results show that this simple filter reduces turnover, from 939% for a monthly rebalanced 12-2 momentum strategy (see Table 1) to 806% for a monthly rebalanced filter 11-1 strategy. This reduction understates the main benefit of this approach. Specifically, the filter ensures that all the stocks in the portfolio remain in the 12-2 momentum strategy in month $t+1$. We can therefore rebalance the portfolio every other month, which results in a 615% turnover, while still maintaining high exposure to momentum. This exposure is evident in the return decay, or lack thereof, of the filter 11-1 strategy. Returns actually increase when rebalanced every other month (15.08%) compared to rebalancing every month (14.27%). The slower rate of decay persists at longer horizons, for example, returns decrease to only 14.48% over 3 months (vs. 13.30% for 12-2 momentum) and to only 12.63% over 6 months (vs. 10.72% for 12-2 momentum). The intuition for this slower decay is that by incorporating information about $t+1$ momentum into our month- t portfolio, we reduce the speed at which the portfolio holdings lose momentum, and delay the onset of the return decay, which begins immediately for 12-2 momentum.

Given the tradeoff momentum investors face between trading fast at a high cost or slowly with return decay, the lower return decay associated with filter 11-1 momentum means that the trader will choose to trade more slowly. Indeed, when we estimate net returns, we find that the filter 11-1 momentum outperforms 12-2 momentum, and that slower trading produces higher returns. For example, rebalancing every 3 months produces the highest net return for filter 11-1 momentum (12.04%), which is 1.15% higher than the net return of 12-2 momentum rebalanced every 3 months. When we consider risk, we find that the portfolio rebalanced every 5 months produces the highest Sharpe ratio of 0.457.¹⁹

¹⁹ Statistical inference on the difference in Sharpe ratios in all the tables is based on a paired block bootstrap procedure with a block length of 12 observations, which preserves the serial dependence structure of the return series. For each bootstrap replication, both return series are resampled jointly, and Sharpe ratios are recomputed to obtain the empirical distribution of their difference. The reported p-values correspond to one-sided tests of whether the Sharpe ratio difference is significantly positive (if the observed difference is positive) or significantly negative (if the observed difference is negative). To facilitate two-sided inference, these one-sided p-values are multiplied by two.

3.1.2 Filter 11-2 Momentum

We next consider a variation on the filter 11-1 strategy that trades off certainty about the momentum status of the portfolio in month $t+1$ against avoiding stocks with high returns in month $t-1$ that are associated with subsequent reversals, and are excluded from the 12-2 momentum rule. Specifically, we filter stocks from the 12-2 momentum portfolio that are not in the 11-2 momentum portfolio at the end of month $t-1$, which we term the *filter 11-2* strategy. Panel B of Table 2 reports the performance of this strategy at different rebalancing horizons. This strategy outperforms 12-2 momentum at every rebalancing horizon. Furthermore, at shorter horizons, by avoiding stocks that are likely to experience reversals, the strategy substantially outperforms the filter 11-1 strategy. For example, it produces 16.62% return with monthly rebalancing (vs. 15.52% for 12-2 momentum and 14.27% for filter 11-1), 16.47% return when rebalanced every two months (vs. 14.79% for 12-2 momentum and 15.08% for filter 11-1), and 15.11% when rebalanced quarterly (vs. 13.30% for 12-2 momentum and 14.48% for filter 11-1).

When we estimate net returns, we find that rebalancing filter 11-2 every two months produces the highest net returns (13.29%), which is larger than the net returns produced by filter 11-1 or 12-2 momentum at any rebalancing frequency. When we account for the volatility of the strategy, we also find that rebalancing filter 11-2 every two months produces the highest Sharpe ratio (0.497), which is larger than the highest Sharpe ratio (0.462) achieved by either the 12-2 momentum or filter 11-1 strategies. Given that this approach produces significantly higher gross returns, net returns, and Sharpe ratios at every rebalancing frequency (except for the one-month horizon for Sharpe ratio), we expect that most investors would choose the filter 11-2 strategy over 12-2 momentum.

3.2 Predictability of Future Momentum beyond Month $t+1$

Investors who can predict the momentum portfolio beyond month $t+1$ should be able to further decrease the return decay and trading costs associated with momentum strategies. Although it is more difficult to predict the constituents of the future momentum portfolios beyond one month, we can still forecast them with some accuracy. For example, consider the momentum portfolio two months ahead. At the end of month $t-1$, 90.9% (10 out of 11 months, $t-10$ to $t-1$) of the return information used to form the $t+2$ momentum portfolio is known. We can use this information to

build portfolios at the end of month $t-1$ that are highly correlated with future momentum portfolios. Given our results for the 11-1 and 11-2 filtered momentum strategies, we would expect these correlated portfolios to experience less return decay than 12-2 momentum, and therefore they would outperform 12-2 momentum at slower rebalancing frequencies.

Panel A of Table 3 presents the correlations among the current and future (from 1 to 11 months ahead) momentum deciles and 12 momentum formation rules: 12-2 momentum, 11-1 momentum, 10-1 momentum, ... , 2-1 momentum, 1-1 momentum. We focus on these rules, because they have the most overlap with 12-2 momentum at each future horizon. For example, in month $t+2$, current 10-1 momentum will become 12-3 momentum, and overlap 90.9% (10 out of 11 months) with future 12-2 momentum. In month $t+6$, current 6-1 momentum will become 12-7 momentum, and overlap 54.5% (6 out of 11 months) with future 12-2 momentum. The approach with maximum overlap can be generalized as the current $(12-k)-1$ momentum portfolio having the highest correlation with 12-2 momentum in month $t+k$.²⁰

As expected, current 11-1 momentum has a correlation of 1.00 with 12-2 momentum in month $t+1$. However, we also find strong predictability beyond month $t+1$. Current 10-1 momentum has a correlation of 0.94 with 12-2 momentum in month $t+2$ (3 months ahead); current 7-1 momentum still has a correlation of 0.78 with 12-2 momentum in month $t+5$ (6 months ahead); and current 2-1 momentum still has a correlation of 0.42 with 12-2 momentum in month $t+10$ (11 months ahead). These correlations are all much higher than the correlation between current 12-2 momentum and future 12-2 momentum. For example, current 12-2 momentum has a 0.79 correlation with 12-2 momentum in month $t+2$, a 0.53 correlation with 12-2 momentum in month $t+5$, and only a 0.12 correlation with 12-2 momentum in month $t+10$. The higher correlations between the maximum overlap portfolios and subsequent 12-2 momentum portfolios compared to the current 12-2 momentum portfolio imply that they will experience lower return decay than 12-2 momentum. However, there is a tradeoff; although these portfolios are more highly correlated with future momentum, they have a lower correlation with current momentum. For example, the 10-1 portfolio has a 0.82 correlation with current momentum, the 7-1 portfolio has a 0.64 correlation with current momentum, and the 2-1 portfolio has only a 0.20 correlation with current

²⁰ There are 78 momentum portfolio formation rules that use the months between $t-12$ and $t-1$ to rank stocks. We find that the maximum overlap portfolios presented in Table 3 achieve the highest correlations with future momentum among all possibilities.

momentum. This lower correlation with current momentum suggests that these portfolios will not perform as well as 12-2 momentum over short horizons.

We also consider correlations with future momentum when we exclude the first month from the formation rule. These results are presented in Panel B of Table 3 and show that by excluding the first month, we reduce our ability to predict future momentum. Specifically, the 11-2 portfolio, which captures 10 of the 11 months used to form the 12-2 momentum portfolio at $t+1$, has a correlation of 0.93 with 12-2 momentum at $t+1$. At $t+5$, 7-2 momentum has a 0.72 correlation with future 12-2 momentum (vs. 0.78 for 7-1 momentum).²¹ However, by excluding month $t-1$ in the formation rule, which is associated with short-term reversals, we achieve higher correlations with current momentum. For example, the 11-2 portfolio has a 0.93 correlation with 12-2 momentum in month t , whereas 11-1 momentum has a 0.88 correlation. As the results in Table 2 show, avoiding the front month in the formation rule increases performance over shorter investment horizons.

3.3 Extended Filtered Strategies

Given the predictability of future momentum beyond one month, we generalize the filtered strategies for slower rebalancing frequencies, with the aim of further improving performance net of trading costs. We draw on the correlation matrix presented in Table 3 and select portfolio formulation rules that maximize the correlation with 12-2 momentum at each investment horizon. We filter out stocks that leave the top and bottom momentum deciles at any horizon, excluding month $t-1$ from the formation rule because of the reversals associated with $t-1$ returns. For example, consider a 3-month horizon, we construct the filter 3 strategy as follows: the long (short) leg includes stocks with returns that are in the top (bottom) decile of returns for the 12-2, 11-2 and 10-2 windows. In other words, if a stock exits the top (bottom) decile for any of the three windows it is filtered out of the portfolio. We generalize this approach through 12 months.

Figure 4 and Table 4 present the returns of the filtered momentum strategies. The returns of filtered strategies are considerably higher than 12-2 momentum at matching rebalancing periods. The highest return, 17.57%, is achieved by the filter 8 strategy, which is 8.60% higher than the 12-2 momentum strategy rebalanced every 8 months, and even 2.05% higher than 12-2

²¹ Additionally, we are no longer able to forecast future momentum at $t+11$ if we exclude month $t-1$, as there is no overlap beyond $t+10$. The table therefore only reports results through $t+10$.

momentum rebalanced monthly. The filter 8 strategy requires 311% annual turnover, which leads to a net return of 16.02%, compared to 10.83% for 12-2 momentum rebalanced monthly, and 11.73% for 12-2 momentum rebalanced every other month, which produces the highest net return of any 12-2 momentum strategy.

One concern is that the restrictive selection process imposed by the filtered strategies, especially at longer horizons, will increase the risk associated with the strategy. For example, to be included in the filter 8 strategy, a stock must be in the top (bottom) decile of returns for all of 12-2, 11-2, 10-2, 9-2, 8-2, 7-2, 6-2, and 5-2 windows. However, this concentration is somewhat offset by the Jegadeesh and Titman (1993) overlapping portfolios approach. This approach increases the breadth and capacity of the strategy while also reducing risk, compared to a non-overlapping portfolios approach. As a result, while the 12-2 momentum portfolio rebalanced monthly holds on average 345 stocks and has a 26.30% standard deviation, the filter 8 strategy, which involves eight overlapping portfolios, holds on average 353 stocks, and has a standard deviation of 26.62%.²²

The increase in return, without a commensurate increase in risk results in the longer horizon filtered approaches having higher Sharpe ratios than 12-2 momentum, as well as the filter 11-1 and filter 11-2 strategies. Specifically, the maximum Sharpe ratio of 0.616 is achieved by the filter 9 strategy. In comparison, the maximum Sharpe ratio achieved by 12-2 momentum, rebalancing every two months, is 0.462, by filter 11-1, rebalancing every 5 months, is 0.457, and by filter 11-2, rebalancing every two months, is 0.497.

3.4 Blended Strategies

We next develop strategies that incorporate information about future momentum in a more measured manner that allows us to continue to build portfolios comprised of the top and bottom deciles of the investment universe most correlated with future momentum. We assume that, for a given rebalancing frequency, a momentum investor will want to maintain a high correlation to momentum over their holding period. Let's again consider a momentum investor with a two-month rebalancing frequency. As shown in Table 3, 12-2 momentum has a correlation of 1.00 with month t momentum, but a correlation of 0.88 with month $t+1$ momentum. 11-2 momentum has a

²² In contrast, if a non-overlapping portfolios approach is used, the filter 8 approach would hold only 96 stocks on average.

correlation of 0.93 with both month t momentum and month $t+1$ momentum. The filter 2 approach holds the subset of stocks that are in the top and bottom deciles of both the 12-2 and 11-2 momentum rankings, and by doing so filters some stocks out of the portfolio and thus decreases the number of holdings. We propose an alternative “blended” approach, which instead assigns 50% weight to the 12-2 return and 50% weight to the 11-2 return, and then sorts stocks into deciles based on the simple average.²³

We generalize this approach for a rebalancing frequency of N months; where the investor assigns $1/N$ weight to each return between the 12-2 return and $(12-N+1)$ -2 return, and we term the strategy the *blended N* strategy. Panel C of Table 3 reports correlations of this approach with subsequent momentum. The results confirm that these blended momentum strategies achieve a more stable correlation with future momentum than current 12-2 momentum, since their average exposure over each horizon is greater.²⁴

We next consider the returns associated with the blended momentum strategies. Figure 4 and Table 5 show that compared with 12-2 momentum, blended momentum strategies significantly improve returns at rebalancing frequencies beyond three months. The blended momentum strategy is designed to reduce the return decay associated with 12-2 momentum and is successful in this goal. For example, the return of 12-2 momentum strategy rebalanced every four months decay 2.95% compared to the return of 12-2 momentum rebalanced monthly (from 15.52% to 12.58%). In contrast, the return of the blended 4 strategy decays by only 1.65% (from 15.52% to 13.87%).

There are at least three benefits to less frequent trading that are realized by the blended strategies: lower trading costs, increased capacity, and reduced risk. Reducing the return decay associated with slow trading makes it more attractive and better enables investors to capture these benefits. For example, the blended 4 strategy requires only 409% annual turnover vs. 939% for

²³ This approach is similar to the optimal strategy proposed by Gârleanu and Pedersen (2013), who develop a theoretically motivated dynamic portfolio policy when trading is costly and returns are predictable. They consider a situation where an investor already knows what some of their signals will look like in the future (like momentum in month $t+1$) and show that the optimal portfolio will be a weighted average of current and past “aim” portfolios. They also draw intuitive real-world analogies to optimal trading behavior: a hunter leading a duck or in the words of Wayne Gretzky to “skate to where the puck is going to be, not where it has been”.

²⁴ As expected, Panel C of Table 3 reports that excluding $t-1$ from the formation rule reduces correlations with future momentum. The correlations of the strategy excluding the front month are lower than when the front month is included. For the two-month horizon, the correlation of the strategy is 0.91 (vs. 0.93 when the front month is included), and this pattern holds for every rebalancing horizon. In unreported results, we find that for all but the longest rebalancing frequencies (10 and 11 months) excluding the front month improves performance as we avoid the short-term reversals associated with $t-1$.

monthly-rebalanced 12-2 momentum. This results in a net return of 11.83%, which is 1.00% larger than the net return of monthly-rebalanced 12-2 momentum, and 1.27% larger than the net return of 12-2 momentum rebalanced every four months. The blended 4 strategy also holds more stocks on average than monthly rebalanced 12-2 momentum (591 vs. 345), which increases the capacity of the strategy for large institutional investors. The greater number of holdings (in addition to diversifying rebalance timing luck) also reduces the risk of the blended 4 strategy, which has a 24.30% standard deviation, compared to a standard deviation of 26.30% for monthly rebalanced 12-2 momentum. This improvement in risk contributes to the blended 4 strategy having the highest Sharpe ratio among the blended strategies, 0.487, which is 0.049 larger (11.1% on a relative basis) than the Sharpe ratio produced by 12-2 momentum rebalanced every four months. It is also 0.024 larger (5.3% on a relative basis) than the 12-2 momentum rebalanced every two months, which has the highest Sharpe ratio among 12-2 momentum strategies.

Although the blended 4 strategy produces the highest Sharpe ratio of any of the blended strategies, we find relatively stable Sharpe ratios using these strategies, as the lower net returns associated with slower rebalancing are offset by lower risk. For example, the blended 6 strategy has a Sharpe ratio is 0.484, the blended 8 strategy has a Sharpe ratio of 0.474, and the blended 10 strategy has a Sharpe ratio of 0.441. These differences are much smaller than the differences in Sharpe ratios for 12-2 momentum at the same rebalancing frequencies: 0.399 for 6 months, 0.351 for 8 months, and 0.288 for 10 months. This result highlights a more subtle benefit to the blended momentum approach, that its performance is more robust to subjective implementation choices related to the rebalancing horizon.

Ultimately, we expect heterogeneity in investors' preferences for the optimal momentum strategy, both in choosing between blended and filtered momentum, as well as choosing the optimal version of the strategy (i.e., the rebalancing frequency). Investors differ in their risk aversion, tracking error considerations, trading efficiency, liquidity preferences, capacity, and confidence in the persistence of momentum returns (among other considerations).²⁵ Which investment strategy is optimal will vary based on each investor's unique characteristics. For example, a large investor with high-capacity requirements may prefer the blended approach, which allows them to target a broader universe of stocks. And small investors with lower capacity needs may choose the filtered approach to maximize the return and Sharpe ratio of their portfolio.

²⁵ In subsequent analysis, we explore how out-of-sample decay impacts design choice in momentum strategies.

However, it is difficult to envision a set of investor characteristics or preferences that would lead an investor to prefer 12-2 momentum to the blended or filtered approaches, given that these strategies dominate on a gross, net, and risk-adjusted return basis. In the remainder of the paper, we present additional tests to better understand the drivers, robustness, and applicability of blended and filtered momentum.

4. Additional Results

4.1 Blended & Filtered UMD vs. Alpha

We next consider if momentum exposure or alpha drives the outperformance of the filtered and blended momentum strategies compared to 12-2 momentum. Given that we explicitly construct the filtered and blended strategies to be correlated with future 12-2 momentum, we expect that these strategies will have higher momentum loadings than the 12-2 momentum strategy rebalanced at the same frequency, and that the premium associated with these loadings drives our results. Panel A of Table 6 presents the up-minus-down (UMD) loadings of 12-2, filtered, and blended momentum over various rebalancing frequencies using the Carhart 4-factor regression.²⁶ We find that the monthly rebalanced 12-2 momentum strategy has a UMD loading of 1.43. However, as expected the UMD exposure of the 12-2 momentum strategy decays rapidly, whereas the UMD exposure of the other strategies stays at a higher level for longer. For example, at a 6-month rebalancing frequency the UMD exposure of 12-2 momentum drops to 1.16, while the exposure of the blended strategy is 1.24 and the exposure of the filtered strategy is even higher at 1.39. The differences in UMD exposures between the three strategies are statistically significant at every horizon.²⁷ We therefore conclude that, at least some of the outperformance of the filtered and blended momentum strategies is the result of these strategies maintaining high exposure to the UMD factor when rebalancing less frequently.

We next turn our attention to the four-factor alpha produced by each strategy, which gauges performance beyond momentum exposure, as well as exposure to market beta, the size factor

²⁶ We use the UMD factor from Kenneth French's data library. This factor is the average return on two high prior return portfolios - a portfolio of small stocks (below median NYSE market equity) and large stocks (above median NYSE market equity) with prior 12-2 return above the 70th NYSE percentiles - minus the average return on two low prior return portfolios - a portfolio of small stocks and large stocks with prior 12-2 return below the 30th NYSE percentiles.

²⁷ For brevity, Table 6 does not report the differences in UMD exposure and Alpha for the blended and filtered strategies. We compute these figures and find that the differences of both measures are statistically significant at the 1% level at every rebalancing horizon.

(SMB), and the value factor (HML). The results, presented in Panel B of Table 6, show that the 12-2 momentum strategy produces positive alpha, which decays as the rebalancing frequency increases. For example, when rebalanced monthly, it produces an annualized alpha of 4.64%; when rebalanced every 6 months, the alpha decreases to 2.22%; and when rebalanced every 11 months, the alpha further decreases to 0.88%.

The blended momentum strategies also produce positive alpha at similar levels to 12-2 momentum. For example, the annualized alpha for blended 6 strategy is 2.76% vs. 2.22% for the 12-2 momentum. However, we do observe that the alpha for blended momentum strategies decay more slowly than the alpha of 12-2 momentum resulting in higher alphas at longer horizons (e.g., 1.63% vs. 0.88% at 11 months) although these differences are only marginally significant at the 8- and 9-month horizons.

We next examine the four-factor alphas generated by the filtered momentum strategies. A couple of interesting patterns emerge. First, the magnitude of the alpha generated by the filtered momentum strategies is substantially larger than the alpha generated by the 12-2 and blended momentum strategies. For example, the alpha on the filter 6 strategy is 6.52%, which is several times larger than the 2.22% (2.76%) alpha for 12-2 (blended) momentum rebalanced every six months. The differences between the alphas of the filtered strategy and the 12-2 and blended momentum strategies are statistically significant at every horizon. Second, whereas the alpha decays for the 12-2 momentum and blended strategies at slower rebalancing frequencies, the alpha of the filtered strategies generally increases at slower rebalancing frequencies. For example, the alpha of the filtered strategy increases from 4.86% for a 3-month rebalancing frequency to 8.43% for a 9-month rebalancing frequency, compared to alpha of 12-2 momentum which declines from 2.98% to 1.46% over the same rebalancing frequencies, and the alpha of the blended strategy declines from 3.11% to 2.48% over the same rebalancing frequencies.

Taken together these results suggest that the outperformance of the filtered strategy and, to a lesser extent, the blended strategy are driven by both increased exposure to momentum and higher alpha.

4.2 Post Publication Performance

When pursuing an active investment strategy an investor must decide if the benefits of that strategy outweigh the costs. Thus far, our analysis has focused on the net benefits to momentum

investing, assuming that they will remain at the level observed in our sample. However, McLean and Pontiff (2016) and Calluzzo et al. (2019) document that after publication anomaly returns decline by approximately 50%. The post-publication decay in returns may lead to different optimal rebalancing horizons for investors trying to maximize their net returns. Therefore, we consider the returns to momentum investing strategies in the period after the publication of the momentum anomaly. Consistent with the literature, we find that the return of 12-2 momentum drops from 21.71% during the sample period of the original study (1965-1989) to 12.66% in the period after the publication of momentum in 1993, a 41.7% relative drop.

Table 7 presents the net returns and Sharpe ratios of the 12-2, blended, and filtered momentum in the post-publication period. We continue to find that the blended and filtered approaches produce superior net returns and Sharpe ratios relative to the 12-2 momentum, and that the filtered approach outperforms the blended approach. Specifically, the maximum net return achieved by 12-2 momentum is 8.68% with monthly rebalancing. In contrast, the blended 2 strategy produces a net return of 10.42% at the 2-month horizon, which is the greatest of any of the blended strategies, and the filter 9 strategy produces a net return of 12.53% at the 9-month horizon, which is the highest of any of the momentum strategies.

We also find that the blended and filtered approaches dominate 12-2 momentum in the post-publication period when we evaluate the strategies using their Sharpe ratios. Specifically, we find that among 12-2 momentum strategies, rebalancing every two months produces the highest Sharpe ratio of 0.297. In contrast, the blended 2 strategy produces a Sharpe ratio of 0.370, which is the greatest of any of the blended strategies, and the filter 9 strategy produces a Sharpe ratio of 0.443, which is the highest of any of the momentum strategies. One interesting observation between the full sample and the post-publication period is that the net returns and Sharpe ratios of the 12-2 and filtered strategies drop by more than those of blended momentum approach (although the filtered approach still produces the highest net returns and Sharpe ratios).²⁸

4.3 Long-Only Momentum

²⁸ Specifically, for the blended approach the highest net return (Sharpe ratio) in the post-publication period drops by 1.49% (0.117) compared to the full sample, whereas it drops by 3.05% (0.165) for 12-2 momentum, and 3.49% (0.173) for the filtered approach.

Many investors are unable or unwilling to take short positions in stocks, which prevents them from investing in the short leg of momentum. For example, some institutional investors are restricted from using short sales (e.g., Calluzzo et al. 2025). Some investors may also have preferences against shorting stocks because of how it changes the risk and return attributes of a portfolio. These constraints and preferences are evidenced in the momentum ETFs marketed to retail investors that only pursue the long leg of momentum (e.g., Broman and Moneta 2024).

Given the prevalence of long-only momentum products, in this section we examine the performance of the long leg of the blended and filtered strategies, compared to the long leg of 12-2 momentum. Table 8 reports the net returns and Sharpe ratios of the long leg of the momentum strategies. Similar to the long-short results, we find that the blended strategy outperforms the 12-2 momentum rebalanced at the same frequency. For example, the long leg of the blended 2 strategy has a net return of 14.22% at the two-month horizon, which is 0.41% higher than the net return of 12-2 momentum strategy rebalanced every two months, and also larger than the 13.64% net return of 12-2 momentum rebalanced every month.²⁹ We find similar results when we measure performance using the Sharpe ratio. The long-only blended approach produces a significantly higher Sharpe ratio than the 12-2 momentum strategy rebalanced at the same frequency except for the two-month horizon. As with the long-short strategies, we find that the long-only filtered approach outperforms both the 12-2 and blended momentum strategies. We find that the largest net return of 16.82% is achieved by the filter 8 approach, which is higher than the highest net returns achieved by the 12-2 and blended approaches. The highest Sharpe ratio of 0.608 is also achieved by the filter 8 approach.

4.4 Crash Risk

We next study potential crash risk associated with the blended and filtered strategies. Daniel and Moskowitz (2016) document the crashes associated with 12-2 momentum, which resulted in a 200% loss from July 1932 to August 1932, and in a 155% loss from March 2009 to May 2009. The tendency of an investment strategy to produce more extreme returns than what is

²⁹ The high net returns of the long-only portfolios (which are larger than those of the long-short portfolios) highlights the tradeoffs investors face when choosing long-only vs. long-short portfolios, and why some may have preferences against shorting stocks. Specifically, including the short leg of momentum decreases total return, but also decreases the correlation of the momentum strategy with the market portfolio (by decreasing the market beta of the portfolio) and increases the CAPM alpha of the strategy.

implied by the volatility under the assumption of a normal distribution can be gauged using the higher order risk measures of skewness and kurtosis. Specifically, skewness measures the asymmetry of a distribution and kurtosis measures the fatness of the distribution's tails (i.e., the occurrence of extreme values vs. what is predicted by a normal distribution).

Table 9 reports the skewness and kurtosis of the momentum strategies. Monthly rebalanced 12-2 momentum has a -1.68 skewness and 14.30 kurtosis, which highlights the risk of extreme losses with 12-2 momentum and is consistent with the existence of momentum crashes. The blended and filtered strategies do not substantially affect the skewness and kurtosis of investment strategies. For example, the skewness of the blended 6 strategy is -1.74 and the kurtosis of the strategy is 14.94 , and the skewness of the filter 6 strategy is -1.78 and the kurtosis of the strategy is 15.36 . These tail risk figures are not economically different from the skewness and kurtosis of monthly rebalanced 12-2 momentum, nor are they economically different from the tail risk metrics for 12-2 momentum rebalanced at the same six-month frequency, which has a skewness of -1.79 and a kurtosis of 14.66 .

We next measure tail risk using the median drawdown size and length of each strategy. These measurements have an intuitive interpretation – how much and for how long an investor is down from their previous high-water mark of the strategies at any given point in time – and also account for the fact that strategies with higher risk-adjusted returns are less likely to experience large and prolonged drawdowns. For 12-2 momentum, we do not find a consistent relation between drawdowns and rebalancing horizon. For example, the median drawdown size (length) of monthly rebalanced 12-2 momentum is 44.97% (79 months), whereas the median drawdown size (length) of 12-2 momentum rebalanced every six months is 45.25% (52.5 months).

When we examine the drawdown size and length of the blended and filtered strategies, we find that these approaches are able to mitigate the severe drawdowns associated with 12-2 momentum. For example, the median drawdown size (length) of blended 3 strategy is 33.76% (39.0 months), whereas for blended 6 strategy it is 29.50% (30.0 months). The improvement for the filtered strategies is particularly notable. For example, the median drawdown size (length) of filter 3 strategy is 26.98% (21.0 months), whereas for filter 6 strategy it is 24.63% (19.0 months).

4.5 Concentrated Blended and 12-2 Momentum

One concern is whether the superior performance of the filtered momentum approach is driven by its concentrated nature rather than its forward-looking approach. In other words, we may be comparing apples to oranges by looking at filtered portfolios, which have more concentrated holdings compared to the 12-2 or blended approaches. To address this concern, we examine strategies that use the 12-2 and blended approaches that rank stocks by their momentum each month, but instead of investing in the top (and bottom) decile based on those ranks, they concentrate into portfolios of only the highest (and lowest) ranked stocks that contain the same number of stocks as in the filtered approach.

Unsurprisingly, this concentrated approach improves the net returns and Sharpe ratios generated by the 12-2 momentum strategies. For example, as reported in Table 10, the highest net return, 13.96%, and Sharpe ratio, 0.520, are achieved by the concentrated 12-2 momentum strategy with a four-month rebalancing horizon, compared to highest net return (11.73%) and Sharpe ratio (0.462) achieved by unconcentrated 12-2 momentum when it is rebalanced every other month. However, these metrics are still considerably below the highest net return achieved by the filter 8 strategy (16.02%) and the highest Sharpe ratio achieved by the filter 9 strategy (0.616). Furthermore, when we compare concentrated 12-2 to the filtered strategy at matching rebalancing horizons, we find that the concentrated 12-2 momentum produces significantly lower net returns beyond 5-month rebalancing horizons.

We also find that concentration improves the net returns and Sharpe ratios of the blended approaches. Specifically, we find that the highest net return of 15.58% is achieved by the concentrated blended 7 strategy, and the highest Sharpe ratio of 0.559 is achieved by the concentrated blended 5 strategy, compared to the maximum 11.90% net return produced by the unconcentrated blended 2 strategy and the maximum 0.487 Sharpe ratio achieved by the unconcentrated blended 4 strategy. Although these figures fall a little short of the maximum net returns and Sharpe ratios achieved by the filtered approach, we do find that at rebalancing horizons between 2 and 7 months, the blended approach produces higher net returns than the filtered approach (although only at a significant level at the 2-month horizon).

These results provide additional evidence that, although the blended and filtered approaches both dominate 12-2 momentum strategies, neither dominates the other. We only observe one rebalancing horizon (2 months) where the net return of the concentrated blended approach is significantly larger than the net return of the filtered approach, and one rebalancing

horizon (11 months) where the net return of the concentrated blended approach is significantly lower than that of the filtered approach. Therefore, in the real world, we expect investors to differ in how they choose to implement momentum strategies, based on their unique characteristics related to risk aversion, tracking error considerations, trading efficiency, liquidity preferences, capacity, and confidence in the persistence of momentum returns.

4.6 International Evidence

We next conduct out-of-sample tests of our trading strategies in a sample of international stocks from 46 countries in developed and emerging markets using the Global Factor Data based on Jensen et al. (2023).³⁰ The sample period for the international stocks is 1988-2024.³¹

We first consider how a 12-2 momentum strategy rebalanced at different horizons performs in the international sample. The results of this analysis are presented in Table 11. Consistent with the findings for the U.S. sample and with Asness et al. (2013), we find that monthly rebalanced 12-2 momentum strategy generates a 8.34% net return and 0.336 Sharpe ratio. Rebalancing 12-2 momentum strategy every other month produces the highest net return (8.76%) and Sharpe ratio (0.374). As with the U.S. sample, we find that the net returns to 12-2 momentum decay if rebalanced at an interval beyond every other month, as the benefits of slower rebalancing (lower trading costs) are more than offset by its disadvantages (reduced momentum exposure).

We next consider the performance of the filtered and blended momentum strategies in the international sample. As with the U.S. sample, we find that the filtered momentum strategy produces positive net returns and Sharpe ratios. The highest net return (10.45%) and Sharpe ratio (0.421) are achieved when the strategy is rebalanced at a two-month horizon. Furthermore, the net return and Sharpe ratio of the filtered strategy are significantly larger than those produced by 12-2 momentum at every horizon (except for the Sharpe ratio at the three-month horizon). We also find that the blended momentum strategy outperforms 12-2 momentum at every horizon.

³⁰ Developed markets include Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, and the UK. Emerging markets include Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Korea, Kuwait, Malaysia, Mexico, Peru, Philippines, Poland, Qatar, Saudi Arabia, South Africa, Taiwan, Thailand, Turkey and United Arab Emirates.

³¹ Consistent with Jensen et al. (2023), we winsorize returns at 0.1% and 99.9% each month to account for potential data errors as the Global Factor Data on WRDS are not winsorized.

Rebalancing the blended momentum strategy at a four-month horizon produces the highest net return (9.66%) and Sharpe ratio (0.433).

The persistence of our results in the international setting suggests that the outperformance potential of the filtered and blended strategies is pervasive, and that investors would benefit by employing them in their portfolio design. Additionally, because trading costs are higher in an international setting (e.g., Domowitz et al. 2001 and Frazzini et al. 2018), it is likely that on a relative basis the benefit of investing in momentum using filtered and blended strategies is larger than in the U.S. setting.

4.7 Machine Learning Approach

In this section we explore whether machine learning techniques that use only past year's return information can provide better out-of-sample performance.³² Gu et al. (2020) show that machine learning methods can provide significant predictive gain by allowing nonlinear predictor interactions missed by other methods, and that neural networks are among the most effective in predicting stock returns.³³ Monthly returns are first transformed into cumulative returns for horizons ranging from 1 to 12 months and standardized using only information from the training set to prevent look-ahead bias. For every month, returns are then converted to rank-based versions (scaled to percentiles between 0 and 1) that serve as target variables for model training. The forecasting design follows a forward-expanding, rolling-window scheme, which updates its training as new data is realized in the post-publication period to replicate realistic investment conditions and prevent look-ahead bias. Specifically, we train a model that uses all possible combinations of past returns (78 combinations in total spanning from $t-1$ to $t-12$) to predict subsequent returns over the next 1 to 12 months corresponding to our investment horizons. We use the pre-publication period to train the model and then examine its performance in the post-publication period.³⁴ For each prediction horizon, we train a feedforward neural network model

³² There would be a benefit to using more information such as volume and accounting data. But to compare with the previous techniques we only use past year's return information.

³³ In unreported analysis we also tried to use a regression trees method, the Extreme Gradient Boosting, but it underperformed the neural network model.

³⁴ In the initial iteration, the training set included all observations from August 1927 through December 1991. The validation window then covered January 1992 through December 1992, and the subsequent test window extended from January 1993 through December 1993. After each iteration, the training set was expanded by one additional year, while the validation and test windows were shifted forward by one year. This process was repeated sequentially until the end of the sample. The design thus produced a series of non-overlapping validation and test sets, while ensuring that temporal ordering was strictly preserved.

consisting of 5 hidden layers of decreasing size (200, 128, 64, 32, and 10 neurons, followed by a hyperbolic tangent activation) and a linear output layer.³⁵

For each horizon and month, we form the long and short legs of the investment strategy based on the decile of stocks with the highest and lowest machine-learning return forecasts. Portfolios are formed using the same approach employed in the blended and filtered strategies (e.g., value-weights and overlapping portfolios). It is important to note that the machine learning approach is not explicitly targeting future momentum, but it is related to momentum in the sense that it is using all possible combinations of returns in the previous twelve months to predict subsequent returns.

Table 12 presents the results of this analysis. The machine learning approach produces higher net returns than 12-2 momentum, but the differences are not statistically significant and likely reflect added noise from the machine learning approach. Both its maximum net return (10.55%) and Sharpe ratio (0.474) are achieved when the target objective is to maximize returns (and rebalance) every two months. These results show that machine learning techniques can help predict returns at short horizons. However, comparing the performance of the machine learning approach to the post-publication performance of the filtered and blended approaches (reported in Table 7) shows that it does not outperform these approaches. These results suggest that the intuitive, simple, and tractable portfolio formation rules that govern the filtered and blended approaches compare favorably to the more sophisticated and opaque techniques such as machine learning. However, these results do not rule out the potential for combining the filtered and blended approaches with more complex and sophisticated techniques to further improve performance for investors that aim to target momentum over longer holding periods.

4.8 Transaction Cost Robustness

Which momentum strategy is optimal for an investor is a function of (among other things) how costly it is for them to execute trades. Throughout the paper we compare strategies using

³⁵ Model training includes an early-stopping mechanism (a technique where the training is automatically stopped if the model's performance on the validation set stops improving for a certain number of training rounds) to prevent overfitting. To enhance predictive robustness, we also implement an ensemble strategy. For each horizon and training window, 10 networks were trained with different random seeds, and their outputs were averaged to produce final predictions. Hyperparameters (e.g., the learning rate, the number of hidden layers in each layer, and batch size) were selected based on the literature. We tried to change hyperparameters and number of hidden layers to test for improvement.

returns net of trading costs, based on the assumption that every 100% roundtrip portfolio turnover costs an investor 0.25%. In this section, we replicate our main result under different trading cost assumptions.

We compute the returns of the 12-2, blended, and filtered momentum strategies, assuming 0.125% trading costs for every 100% roundtrip turnover, 0.375% trading costs for every 100% roundtrip turnover, and using the effective bid-ask spread measure proposed by Hasbrouck (2009) and adopted by Novy-Marx and Velikov (2016).³⁶ Table 13 reports the difference in net returns for the blended and filtered momentum strategies compared to 12-2 momentum rebalanced at the same horizon under different trading cost assumptions. For comparison, it also includes figures assuming 0.25% trading costs for every 100% roundtrip turnover, which is used in our main analysis. The results highlight the robustness of our main result: the outperformance of the blended and filtered strategies compared to 12-2 momentum persists at a similar magnitude and level of significance regardless of the level or methodology of assumed trading costs. The difference in returns compared to 12-2 momentum continues to be the greatest at longer horizons where the 12-2 momentum strategy is more susceptible to return decay.

In untabulated analysis we compute breakeven transaction costs for our momentum strategies to better understand the level of costs that would make an investor select our proposed strategies over 12-2 momentum rebalanced monthly. The breakeven cost is defined as the level of annualized roundtrip transaction costs above which an investor would choose the strategy over 12-2 momentum rebalanced monthly with respect to returns and Sharpe ratios. For the filtered momentum strategies, the breakeven trading costs (with one exception) are negative, meaning that even an investor with zero trading costs would choose the filtered momentum strategies. For the blended strategies, we find that an investor needs non-zero trading costs to choose them over 12-2 momentum rebalanced monthly. For example, to choose the blended 2 strategy, an investor would need trading costs above 0.18% (0.07%) for it to produce higher returns (Sharpe ratios) than the monthly rebalanced 12-2 momentum; and to choose the blended 4 strategy the breakeven cost is 0.31% (0.10%). These results provide additional nuance to our main results and highlight that

³⁶ Hasbrouck (2009) uses a Bayesian approach to estimate the cost measure and obtain effective spreads that have a 96.5% correlation with the ones estimated using actual trades from the trade and quote (TAQ) dataset. For missing transaction cost observations, we follow Novy-Marx and Velikov (2016) and use the estimated cost of trading from the nearest stock match (based on size and idiosyncratic volatility) for which a direct trading cost estimate is available.

our results are robust to even very low estimates of trading costs, and thus relevant even for investors with strong execution abilities.

4.9 Additional Tests

In this section, we examine the performance of the filtered and blended strategies in three additional settings that may more closely represent the preferences and/or constraints of practitioners. Specifically, we consider (1) an expanded momentum universe that goes long stocks in the top three momentum deciles and shorts stocks in the bottom three momentum deciles; (2) a sample that includes financials and utilities, which are excluded from our main analysis; and (3) a sample that excludes stocks below the NYSE 30th percentile market capitalization cutoff, which are small and may not be practical for investors to trade.³⁷ These tests help establish that our main results are not driven by the specific implementation choices made in our primary analysis, and that the benefits of the filtered and blended approaches generalize to settings that may more closely mimic how real-world investors target momentum.

We first consider a broader momentum universe. While our main analysis follows the convention of forming long-short portfolios from the top and bottom deciles of the momentum distribution which is common in the academic literature (e.g., Jegadeesh and Titman 1993, Asness 1994), many practitioners (and academic papers, e.g., Carhart 1997) use wider sorts to increase the breadth and capacity of their strategies. Internet Appendix Table 1 reports the net returns and Sharpe ratios of the 12-2, filtered, and blended momentum strategies when we extend the long and short legs to include the top and bottom three deciles. The broader sort mechanically reduces the magnitude of momentum returns, as it dilutes the strongest winners and weakest losers with stocks that have more moderate momentum. For example, the net return of 12-2 momentum rebalanced monthly across the top/bottom 30 percent is 4.91%, compared to 10.83% for the top/bottom decile. Despite the lower return levels, we continue to find that the filtered and blended strategies outperform 12-2 momentum. For example, the filter 8 strategy in this setting produces a 4.01% higher return than 12-2 momentum rebalanced every eight months (7.71% vs. 3.70%), and a 0.176 greater Sharpe ratio (0.423 vs. 0.247). Both differences are statistically significant at the 1% level, and more broadly filtered strategies significantly outperform 12-2 momentum at every rebalancing horizon. The blended strategies more modestly outperform 12-2 momentum in this setting. For

³⁷ Using an NYSE cutoff was suggested by the CIO of a fund company that implements momentum strategies.

example, the blended 8 strategy has a net return of 4.71% (1.02% higher than 12-2 momentum) and a Sharpe ratio of 0.325 (0.077 greater than 12-2 momentum), and the differences in net returns only become statistically significant at rebalancing horizons of six months and beyond. Taken together, these findings are consistent with our main analysis and suggest that real-world investors who use broader momentum sorts would benefit from adopting the filtered or blended formation rules.

Next, we restrict our sample to stocks above the NYSE 30th percentile market capitalization cutoff, which excludes the smallest and least liquid stocks. This test is motivated by the concern that momentum returns, and the benefits of our strategies, may be concentrated in small-cap stocks that are costly to trade, have limited capacity, and therefore may not be practical for many institutional investors. Internet Appendix Table 2 presents the results. As expected, excluding the smallest stocks reduces the returns of all momentum strategies. For example, the net return of 12-2 momentum rebalanced monthly is 8.60%, compared to 10.83% in our main analysis. Nevertheless, the filtered and blended strategies continue to outperform 12-2 momentum. Specifically, the net return (and Sharpe ratio) of the filtered momentum strategy is significantly higher than that of 12-2 momentum at every rebalancing horizon (at the three-month rebalancing horizon and beyond). As with our main results, the outperformance of the blended strategy is more modest, but it also produces significantly greater net returns and Sharpe ratios than 12-2 momentum at the three-month rebalancing horizon and beyond. These findings indicate that the benefits of the filtered and blended approaches are not driven by small, illiquid stocks, and that investors who restrict their universe to more liquid names would still benefit from adopting our formation rules.

Lastly, we examine the impact of including financials and utilities (which are excluded from our main analysis as in much of the academic literature). Internet Appendix Table 3 reports the results. Including these industries does not lead to an economically meaningful impact on the returns for the three momentum strategies, although the net Sharpe ratios increase modestly consistent with improved diversification from their inclusion (e.g. the net Sharpe ratio for the filtered 8 strategy increases from 0.602 to 0.615, and that of blended 8 strategy increases from 0.474 to 0.485). Within this sample, we again find that filtered strategies produce significantly higher net returns and Sharpe ratios than 12-2 momentum at every horizon. The results are more modest for the blended strategies, which produce significantly higher net returns and Sharpe ratios

than 12-2 momentum at the five-month rebalancing horizon and beyond. These results demonstrate that our findings are not an artifact of excluding specific industries and that the filtered and blended approaches continue to generate economically and statistically significant improvements in a more complete investment universe.

Taken together, the results in this section show that the benefits of the filtered and blended momentum strategies generalize across a range of settings that reflect common practitioner implementation choices: broader momentum sorts, restriction to larger, more liquid stocks, and the inclusion of all industries.

5. Conclusion

In the academic literature, the preferred momentum strategy is the 12-2 momentum, which uses the past year's return skipping the most recent month. Returns on momentum strategies are also usually measured over a one-month window and the portfolio is rebalanced every month. However, in practice, investors who target momentum face a tradeoff between high trading costs due to frequent rebalancing and a decay in momentum returns associated with longer holding periods. Therefore, when using longer holding periods, the 12-2 momentum may not provide the best performance after accounting for transaction costs.

This paper proposes strategies for longer holding periods that exhibit less turnover than the 12-2 momentum rebalanced monthly without sacrificing performance. Specifically, we develop two sets of portfolio formation rules that exploit predictability in future momentum, the filtered and blended approaches. The filtered approach creates concentrated portfolios that eliminate stocks that are likely to exit the momentum portfolio in the future. We show that these portfolios outperform the traditional 12-2 momentum on a gross, net, and risk-adjusted basis in both a sample of U.S. stocks and a sample of international stocks. The blended approach weights different momentum strategies based on the holding period, while maintaining the same breadth as the 12-2 momentum. Although this approach does not produce returns as high as the filtered approach, it produces higher net and risk-adjusted returns than the 12-2 momentum, while maintaining high capacity from the large breadth of stocks that it holds. Furthermore, when we modify the blended approach to match the concentration of the filtered approach, we find that it achieves similar levels of risk-adjusted returns as the filtered approach.

Given the heterogeneity among investors in trading costs, capacity, liquidity demands, and expectations about future momentum performance, investors may rebalance their portfolios at different frequencies, and thus have diverse preferences for the exact momentum strategy to implement. Nevertheless, our results show that all investors can benefit by adjusting their momentum formation rules to their holding periods when they target the momentum factor. Finally, these strategies would also be applicable to any signal with fast return decay and that is to some extent predictable.

References

- Antoniou C, Doukas JA, Subrahmanyam A., 2013. Cognitive Dissonance, Sentiment, and Momentum. *Journal of Financial and Quantitative Analysis*. 48(1), pp. 245-275.
- Arnott, R.D., Kalesnik, V. and Linnainmaa, J.T., 2023. Factor momentum. *The Review of Financial Studies*, 36(8), pp.3034-3070.
- Asness, C. (1994), "Variables that Explain Stock Returns," Ph.D. Dissertation, University of Chicago
- Asness, C.S., Moskowitz, T.J. and Pedersen, L.H., 2013. Value and momentum everywhere. *The Journal of Finance*, 68(3), pp.929-985.
- Barberis, N., Shleifer, A. and Vishny, R., 1998. A model of investor sentiment. *Journal of Financial Economics*, 49(3), pp.307-343.
- Berk, J.B., Green, R.C. and Naik, V., 1999. Optimal investment, growth options, and security returns. *The Journal of Finance*, 54(5), pp.1553-1607.
- Broman, M.S. and Moneta, F., 2024. On the anomaly tilts of factor funds. *Financial Management*, 53, pp. 605-635.
- Calluzzo, P., Moneta, F. and Topaloglu, S., 2019. When anomalies are publicized broadly, do institutions trade accordingly? *Management Science*, 65(10), pp.4555-4574.
- Calluzzo, P., Moneta, F. and Topaloglu, S., 2025. Complex Instruments Have Increased Risk and Reduced Performance at Mutual Funds. *Critical Finance Review*, 14(1), pp. 27-64.
- Carhart, M.M., 1997. On persistence in mutual fund performance. *The Journal of Finance*, 52(1), pp.57-82.
- Chu, Y., Hirshleifer, D. and Ma, L., 2020. The causal effect of limits to arbitrage on asset pricing anomalies. *The Journal of Finance*, 75(5), pp.2631-2672.
- Da, Z., Gurun, U.G. and Warachka, M., 2014. Frog in the pan: Continuous information and momentum. *The Review of Financial Studies*, 27(7), pp.2171-2218.
- Daniel, K. and Moskowitz, T.J., 2016. Momentum crashes. *Journal of Financial Economics*, 122(2), pp.221-247.
- Domowitz, I., Glen, J. and Madhavan, A., 2001. Liquidity, volatility and equity trading costs across countries and over time. *International Finance*, 4(2), pp.221-255.
- Ehsani, S. and Linnainmaa, J.T., 2022. Factor momentum and the momentum factor. *The Journal of Finance*, 77(3), pp.1877-1919.

- Frazzini, A., 2006. The disposition effect and underreaction to news. *The Journal of Finance*, 61(4), pp.2017-2046.
- Frazzini, A., Israel, R. and Moskowitz, T.J., 2018. Trading costs. *Available at SSRN 3229719*.
- Gârleanu, N. and Pedersen, L.H., 2013. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6), pp.2309-2340.
- George, T.J. and Hwang, C.Y., 2004. The 52-week high and momentum investing. *The Journal of Finance*, 59(5), pp.2145-2176.
- Grinblatt, M. and Han, B., 2005. Prospect theory, mental accounting, and momentum. *Journal of Financial Economics*, 78(2), pp.311-339.
- Gu, S., Kelly, B. and Xiu, D., 2020. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5), pp.2223-2273.
- Han, Y., Zhou, G. and Zhu, Y., 2016. A trend factor: Any economic gains from using information over investment horizons? *Journal of Financial Economics*, 122(2), pp.352-375.
- Hasbrouck, J., 2009. Trading costs and returns for US equities: Estimating effective costs from daily data. *The Journal of Finance*, 64(3), pp.1445-1477.
- Hoffstein, C., Faber, N., & Braun, S., 2020. Rebalance timing luck: the (dumb) luck of smart beta. Working Paper. Available at SSRN 3673910.
- Hong, H. and Stein, J.C., 1999. A unified theory of underreaction, momentum trading, and overreaction in asset markets. *The Journal of Finance*, 54(6), pp.2143-2184.
- Hou, K., Xue, C. and Zhang, L., 2015. Digesting anomalies: An investment approach. *The Review of Financial Studies*, 28(3), pp.650-705.
- Jegadeesh, N., 1990. Evidence of predictable behavior of security returns. *The Journal of Finance*, 45(3), pp.881-898.
- Jegadeesh, N. and Titman, S., 1993. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1), pp.65-91.
- Jegadeesh, N. and Titman, S., 2001. Profitability of momentum strategies: An evaluation of alternative explanations. *The Journal of Finance*, 56(2), pp.699-720.
- Jensen, T.I., Kelly, B. and Pedersen, L.H., 2023. Is there a replication crisis in finance? *The Journal of Finance*, 78(5), pp.2465-2518.
- Johnson, T.C., 2002. Rational momentum effects. *The Journal of Finance*, 57(2), pp.585-608.

Lesmond, D.A., Schill, M.J. and Zhou, C., 2004. The illusory nature of momentum profits. *Journal of Financial Economics*, 71(2), pp.349-380.

Lettau, M., Ludvigson, S.C. and Manoel, P., 2021. Characteristics of mutual fund portfolios: Where are the value funds? (No. w25381). National Bureau of Economic Research.

Lou, D., 2012. A flow-based explanation for return predictability. *The Review of Financial Studies*, 25(12), pp.3457-3489.

Luo, P., Ravina, E., Sammon, M.C. and Viceira, L.M., 2025. Retail investors' contrarian behavior around news, attention, and the momentum effect (No. w34086). National Bureau of Economic Research.

McLean, R.D. and Pontiff, J., 2016. Does academic research destroy stock return predictability? *The Journal of Finance*, 71(1), pp.5-32.

Novy-Marx, R., 2012. Is momentum really momentum? *Journal of Financial Economics*, 103(3), pp.429-453.

Novy-Marx, R. and Velikov, M., 2016. A taxonomy of anomalies and their trading costs. *The Review of Financial Studies*, 29(1), pp.104-147.

Pagano, M.S., Sedunov, J. and Velthuis, R., 2021. How did retail investors respond to the COVID-19 pandemic? The effect of Robinhood brokerage customers on market quality. *Finance Research Letters*, 43, p.101946.

Patton, A.J. and Weller, B.M., 2020. What you see is not what you get: The costs of trading market anomalies. *Journal of Financial Economics*, 137(2), pp.515-549.

Sadka, R., 2006. Momentum and post-earnings-announcement drift anomalies: The role of liquidity risk. *Journal of Financial Economics*, 80(2), pp.309-349.

Schmidt, P.S., Schrimpf, A., Von Arx, U., Wagner, A.F. and Ziegler, A., 2017. Size and momentum profitability in international stock markets.

Stambaugh, R.F., Yu, J. and Yuan, Y., 2012. The short of it: Investor sentiment and anomalies. *Journal of Financial Economics*, 104(2), pp.288-302.

Subrahmanyam, A., 2018. Equity market momentum: A synthesis of the literature and suggestions for future work. *Pacific-Basin Finance Journal*, 51, pp.291-296.

Vayanos, D. and Woolley, P., 2013. An institutional theory of momentum and reversal. *The Review of Financial Studies*, 26(5), pp.1087-1145.

Wiest, T., 2023. Momentum: what do we know 30 years after Jegadeesh and Titman's seminal paper? *Financial Markets and Portfolio Management*, 37(1), pp.95-114.

Figure 1

This figure plots the annualized gross returns (bars) and portfolio turnover (line) for the 12-2 momentum strategy over various rebalancing frequencies. The left axis indicates the return, and the right axis indicates the portfolio turnover.

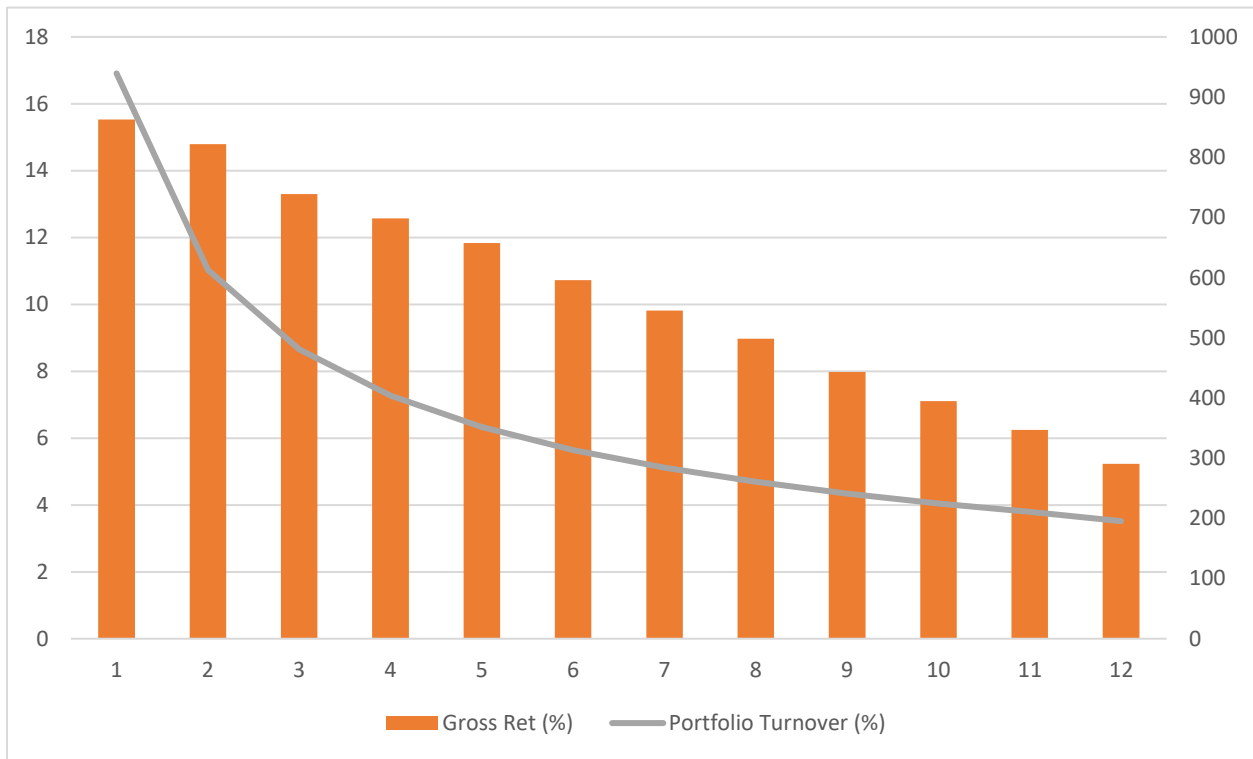


Figure 2

This figure plots the annualized gross returns for the filter 11-1 and filter 11-2 momentum strategies over various rebalancing frequencies. The annualized gross return for the 12-2 momentum strategy is also included for comparison.

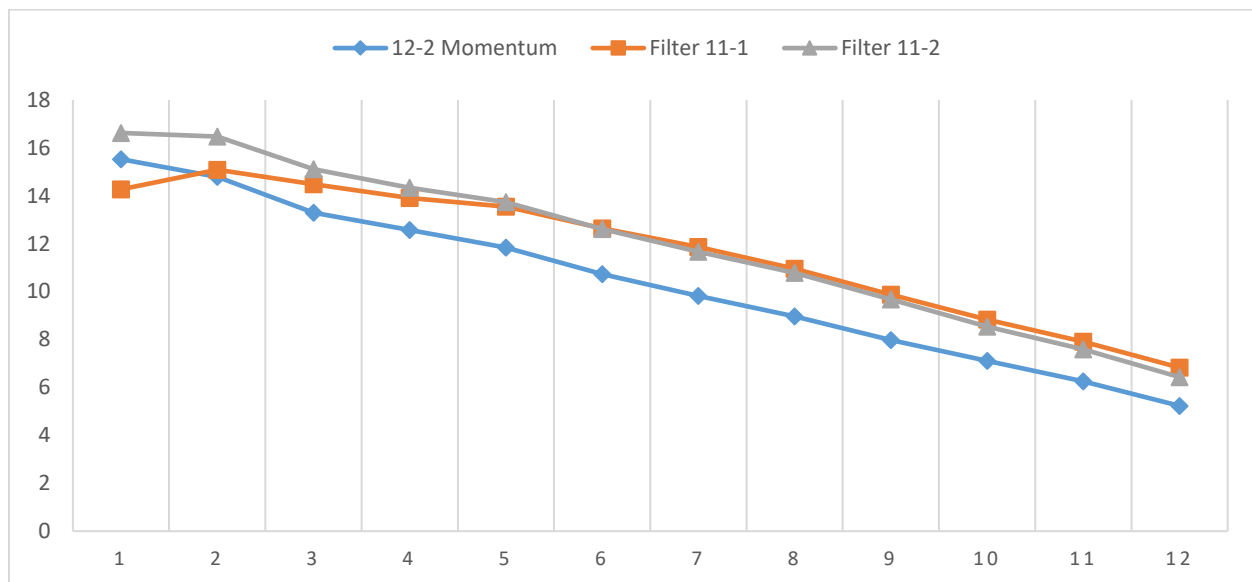


Figure 3

This figure plots the growth of \$1 invested in 12-2 momentum rebalanced monthly, and the filter 9 and blended 4 strategies over our sample period. Returns are computed net of trading costs.

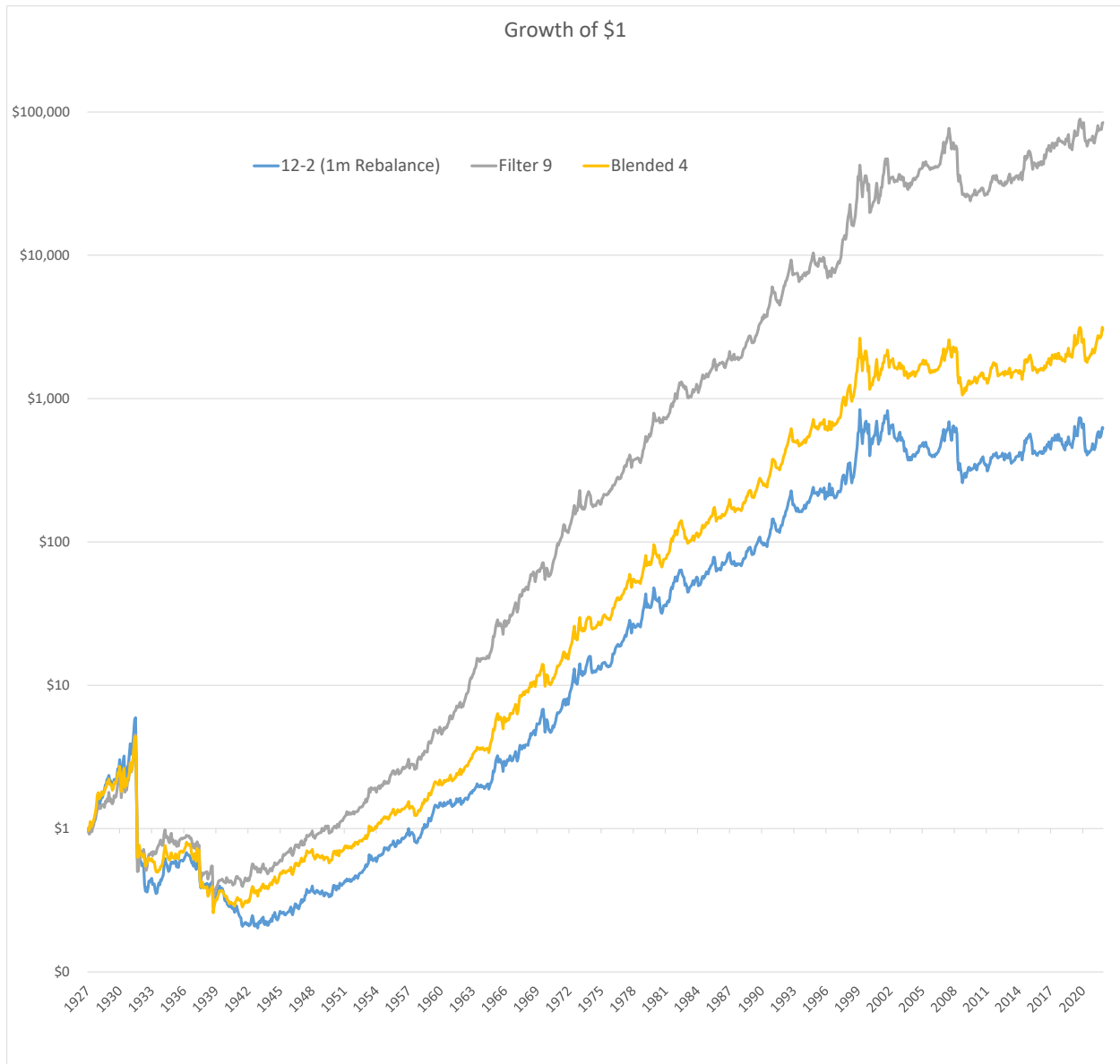


Figure 4

This figure plots the annualized net returns and Sharpe ratios for the blended and filtered momentum strategies over various rebalancing frequencies. The annualized net returns and Sharpe ratios for the 12-2 momentum strategy are also included for comparison.

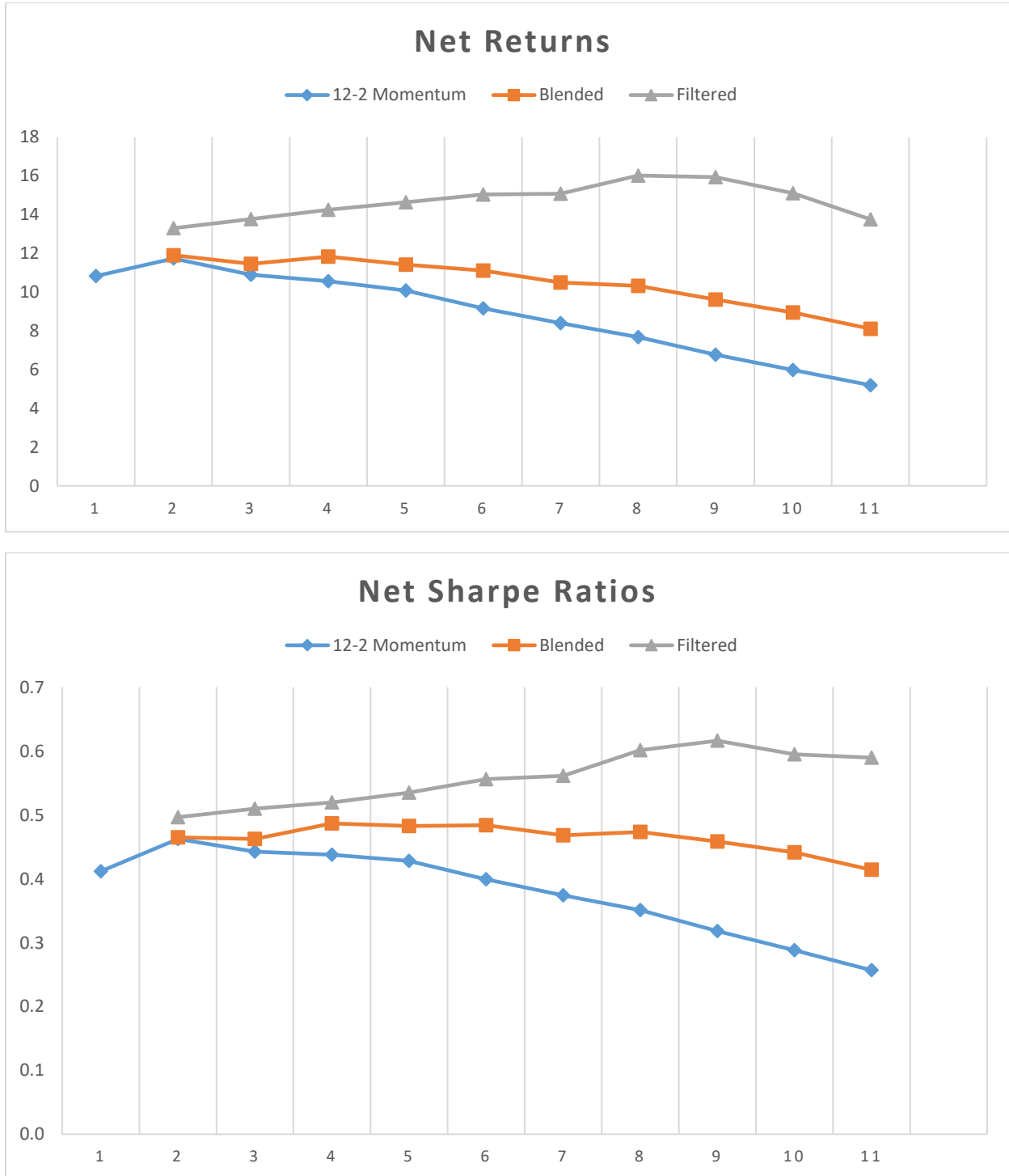


Table 1. Return and Turnover Characteristics for 12-2 Momentum

This table reports the number of holdings, portfolio turnover, annualized gross return, return net of trading costs, standard deviation (SD), and net Sharpe ratio for the 12-2 momentum strategy over various rebalancing frequencies. The horizon column indicates the rebalancing frequency. For example, a horizon of 5 shows results for 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for returns) at the 10, 5, and 1 percent levels.

Horizon	Holdings	Turnover (%)	Gross Ret (%)	Net Return (%)	Net SD (%)	Net Sharpe
1	345	939	15.52***	10.83***	26.30	0.412
2	452	613	14.79***	11.73***	25.37	0.462
3	537	481	13.30***	10.89***	24.61	0.443
4	613	404	12.58***	10.56***	24.11	0.438
5	682	352	11.83***	10.08***	23.55	0.428
6	746	313	10.72***	9.15***	22.93	0.399
7	806	284	9.82***	8.40***	22.45	0.374
8	862	261	8.97***	7.67***	21.86	0.351
9	915	241	7.97***	6.77***	21.29	0.318
10	965	225	7.10***	5.98***	20.76	0.288
11	1012	211	6.25***	5.19**	20.22	0.257
12	1056	195	5.23***	4.25**	19.72	0.216

Table 2. Return and Turnover Characteristics for Filter 11-1 and Filter 11-2 Momentum

This table reports the number of holdings, portfolio turnover, annualized gross return, return net of trading costs, standard deviation (SD), and net Sharpe ratio for the filter 11-1 (Panel A) and filter 11-2 (Panel B) momentum strategies over various rebalancing frequencies. It also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Panel A: Filter 11-1 Momentum

Horizon	Holdings	Turnover (%)	Gross Ret (%)		Net Return (%)		Net Sharpe		Net SD (%)
			Level	Diff	Level	Diff	Level	Diff	
1	249	806	14.27	-1.26	10.24	-0.59	0.353	-0.059*	29.02
2	314	615	15.08	0.29	12.01	0.28	0.429	-0.033	27.97
3	376	489	14.48	1.19**	12.04	1.15**	0.446	0.003	27.02
4	432	415	13.90	1.33***	11.83	1.27**	0.449	0.011	26.35
5	484	362	13.55	1.72***	11.74	1.66***	0.457	0.029**	25.70
6	533	323	12.63	1.91***	11.02	1.86***	0.441	0.042***	24.97
7	581	293	11.86	2.04***	10.39	2.00***	0.427	0.053***	24.34
8	626	269	10.96	1.99***	9.61	1.94***	0.406	0.055***	23.70
9	670	249	9.88	1.90***	8.63	1.86***	0.374	0.056***	23.10
10	712	232	8.84	1.73***	7.68	1.70***	0.341	0.053***	22.51
11	752	215	7.91	1.66***	6.83	1.64***	0.312	0.055***	21.92
12	790	201	6.83	1.60***	5.82	1.57***	0.273	0.057***	21.34

Panel B: Filter 11-2 Momentum

Horizon	Holdings	Turnover (%)	Gross Ret (%)		Net Return (%)		Net Sharpe		Net SD (%)
			Level	Diff	Level	Diff	Level	Diff	
1	277	939	16.62	1.10*	11.93	1.10*	0.430	0.018	27.74
2	363	636	16.47	1.68***	13.29	1.56***	0.497	0.034**	26.78
3	435	500	15.11	1.81***	12.61	1.72***	0.484	0.042***	26.04
4	500	419	14.34	1.76***	12.24	1.69***	0.481	0.043***	25.47
5	560	365	13.74	1.90***	11.91	1.83***	0.480	0.052***	24.82
6	617	326	12.61	1.89***	10.98	1.83***	0.456	0.057***	24.06
7	670	295	11.66	1.84***	10.19	1.79***	0.434	0.060***	23.48
8	721	270	10.78	1.81***	9.43	1.76***	0.413	0.062***	22.84
9	770	249	9.67	1.69***	8.42	1.65***	0.379	0.061***	22.21
10	816	232	8.54	1.43***	7.38	1.40***	0.341	0.053***	21.60
11	860	215	7.59	1.35***	6.52	1.32***	0.310	0.053***	21.02
12	902	199	6.43	1.20***	5.44	1.18***	0.266	0.051***	20.41

Table 3. Correlation of Momentum Strategies with Current and Future 12-2 Momentum Portfolios

This table reports the correlations of various momentum strategies with current (month t) and future 12-2 momentum portfolios. ‘Beg.’ refers to the first month used in the momentum rule and ‘end’ refers to the last month in the momentum rule. For example, beg.=10 and end.=1 indicates the 10-1 momentum. In Panel B, we exclude month $t-1$ from the formation rule. In Panel C, we report average correlations of current 12-2 momentum and two other strategies with 12-2 momentum from month t to month $t+k$. (12- k)-1 strategy assigns $1/(k+1)$ weight each to 12-2, 11-1, 10-1, ..., and (12- k)-1 momentum. (12- k)-2 strategy assigns $1/(k+1)$ weight each to 12-2, 11-2, 10-2, ..., and (12- k)-2 momentum. (12- k)-2 strategy is the same as the blended $k+1$ strategy.

Momentum Rule			Month											
	Beg.	End	t	t+1	t+2	t+3	t+4	t+5	t+6	t+7	t+8	t+9	t+10	t+11
Panel A	12	2	1.00	0.88	0.79	0.70	0.62	0.53	0.45	0.37	0.29	0.20	0.12	0.03
	11	1	0.88	1.00	0.88	0.79	0.71	0.62	0.54	0.46	0.37	0.29	0.21	0.12
	10	1	0.82	0.93	0.94	0.83	0.74	0.65	0.56	0.48	0.39	0.31	0.22	0.13
	9	1	0.77	0.87	0.87	0.89	0.78	0.68	0.59	0.50	0.41	0.32	0.23	0.14
	8	1	0.71	0.82	0.81	0.82	0.84	0.72	0.62	0.53	0.43	0.34	0.25	0.15
	7	1	0.64	0.76	0.76	0.76	0.77	0.78	0.66	0.56	0.46	0.36	0.26	0.16
	6	1	0.58	0.70	0.70	0.70	0.70	0.71	0.72	0.60	0.49	0.38	0.28	0.17
	5	1	0.51	0.64	0.63	0.63	0.64	0.64	0.65	0.66	0.53	0.41	0.29	0.17
	4	1	0.42	0.56	0.56	0.56	0.56	0.57	0.57	0.58	0.59	0.44	0.32	0.19
	3	1	0.33	0.49	0.48	0.48	0.48	0.48	0.49	0.49	0.50	0.51	0.35	0.20
	2	1	0.20	0.39	0.38	0.38	0.39	0.39	0.39	0.39	0.39	0.40	0.42	0.22
	1	1	0.01	0.27	0.26	0.26	0.26	0.26	0.26	0.26	0.27	0.27	0.27	0.30
Panel B	12	2	1.00	0.88	0.79	0.70	0.62	0.53	0.45	0.37	0.29	0.20	0.12	
	11	2	0.93	0.93	0.83	0.74	0.65	0.56	0.47	0.39	0.30	0.22	0.13	
	10	2	0.87	0.87	0.88	0.77	0.68	0.59	0.50	0.41	0.32	0.23	0.14	
	9	2	0.82	0.81	0.82	0.83	0.72	0.62	0.53	0.43	0.34	0.24	0.15	
	8	2	0.76	0.76	0.76	0.76	0.78	0.66	0.56	0.46	0.36	0.26	0.15	
	7	2	0.70	0.69	0.70	0.70	0.71	0.72	0.60	0.49	0.38	0.27	0.16	
	6	2	0.64	0.63	0.63	0.64	0.64	0.64	0.66	0.52	0.41	0.29	0.17	
	5	2	0.57	0.56	0.56	0.56	0.56	0.57	0.57	0.59	0.44	0.31	0.18	
	4	2	0.49	0.48	0.48	0.48	0.48	0.49	0.49	0.49	0.51	0.34	0.20	
	3	2	0.39	0.38	0.38	0.38	0.38	0.39	0.39	0.39	0.39	0.42	0.22	
	2	2	0.27	0.26	0.26	0.26	0.26	0.26	0.26	0.27	0.26	0.27	0.29	
	Panel C			t+1	t+2	t+3	t+4	t+5	t+6	t+7	t+8	t+9	t+10	t+11
12-2 Avg. Correlation			0.94	0.89	0.84	0.80	0.75	0.71	0.67	0.63	0.58	0.54	0.50	
(12-k)-1 Avg. Correlation			0.96	0.93	0.90	0.87	0.84	0.81	0.77	0.74	0.70	0.66	0.62	
(12-k)-2 Avg. Correlation			0.95	0.91	0.88	0.84	0.80	0.77	0.73	0.69	0.65	0.60		

Table 4. Return and Turnover Characteristics for Filtered Momentum

This table reports the number of holdings, portfolio turnover, annualized gross return, return net of trading costs, standard deviation (SD), and net Sharpe ratio for the filtered momentum strategies over various rebalancing frequencies. It also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the filtered momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the filter 5 momentum strategy rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Holdings	Turnover (%)	Gross Ret (%)		Net Return (%)		Net Sharpe		Net SD (%)
			Level	Diff	Level	Diff	Level	Diff	
2	363	636	16.47	1.68***	13.29	1.56***	0.497	0.034**	26.78
3	369	518	16.36	3.06***	13.77	2.88***	0.510	0.067***	27.01
4	371	452	16.51	3.93***	14.25	3.69***	0.519	0.082***	27.43
5	371	410	16.68	4.85***	14.63	4.56***	0.535	0.107***	27.36
6	367	379	16.93	6.21***	15.04	5.88***	0.556	0.157***	27.05
7	362	341	16.78	6.96***	15.07	6.67***	0.561	0.187***	26.84
8	353	311	17.57	8.60***	16.02	8.35***	0.602	0.251***	26.62
9	340	289	17.38	9.40***	15.93	9.16***	0.616	0.298***	25.85
10	321	274	16.47	9.37***	15.10	9.12***	0.595	0.307***	25.38
11	287	265	15.08	8.84***	13.76	8.56***	0.590	0.333***	23.34

Table 5. Return and Turnover Characteristics for Blended Momentum

This table reports the number of holdings, portfolio turnover, annualized gross return, return net of trading costs, standard deviation (SD), and net Sharpe ratio for the blended momentum strategies over various rebalancing frequencies. It also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the blended momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 momentum strategy rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Holdings	Turnover (%)	Gross Ret (%)		Net Return (%)		Net Sharpe		Net SD (%)
			Level	Diff	Level	Diff	Level	Diff	
2	442	600	14.90	0.11	11.90	0.17	0.465	0.003	25.59
3	519	478	13.85	0.55	11.46	0.57	0.462	0.020	24.79
4	591	409	13.87	1.30***	11.83	1.27***	0.487	0.049**	24.30
5	661	364	13.24	1.40***	11.41	1.34**	0.483	0.055**	23.65
6	731	333	12.78	2.06***	11.11	1.96***	0.484	0.085***	22.97
7	803	302	12.00	2.19***	10.49	2.10***	0.468	0.094***	22.41
8	877	275	11.69	2.72***	10.32	2.65***	0.474	0.123***	21.79
9	953	251	10.87	2.90***	9.61	2.85***	0.458	0.140***	20.98
10	1028	231	10.10	3.00***	8.95	2.96***	0.441	0.153***	20.26
11	1100	214	9.18	2.93***	8.11	2.91***	0.414	0.157***	19.58

Table 6. UMD and Alpha Estimates of Momentum Strategies

This table reports the UMD beta (Panel A) and annualized alpha (Panel B) estimates from Carhart 4-factor regressions for the filtered and blended momentum strategies. For comparison, it also includes the estimates and differences when using as dependent variable the returns of 12-2 momentum at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 and filter 5 momentum strategies rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. */**/** denotes statistical significance (reported only for differences) at the 10, 5, and 1 percent levels.

Horizon	Panel A: UMD Beta					Panel B: 4-Factor Alpha				
	12-2 Level	Filtered Level	Diff	Blended Level	Diff	12-2 Level	Filtered Level	Diff	Blended Level	Diff
1	1.43					4.64				
2	1.39	1.46	0.07***	1.41	0.02**	4.06	5.13	1.07**	3.85	-0.21
3	1.33	1.47	0.13***	1.36	0.03***	2.98	4.86	1.88***	3.11	0.13
4	1.28	1.47	0.19***	1.33	0.05***	2.84	5.16	2.32***	3.33	0.49
5	1.22	1.43	0.21***	1.28	0.06***	2.72	5.79	3.07***	3.05	0.33
6	1.16	1.39	0.23***	1.24	0.08***	2.22	6.52	4.29***	2.76	0.54
7	1.10	1.34	0.24***	1.19	0.09***	1.90	6.89	4.99***	2.44	0.54
8	1.04	1.32	0.28***	1.14	0.10***	1.71	8.20	6.49***	2.61	0.90*
9	0.97	1.26	0.29***	1.08	0.11***	1.46	8.43	6.97***	2.48	1.02*
10	0.91	1.20	0.29***	1.03	0.12***	1.21	7.98	6.77***	2.03	0.82
11	0.86	1.07	0.21***	0.97	0.11***	0.88	7.12	6.24***	1.63	0.75

Table 7. Post-Publication Results

This table reports the net returns and net Sharpe ratios for the 12-2, blended, and filtered momentum strategies in the period after the 1993 publication of Jegadeesh and Titman's paper on momentum. It also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 and filter 5 momentum strategies rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Net Return (%)					Net Sharpe				
	12-2		Filtered		Blended		12-2		Filtered	
	Level	Level	Diff	Level	Diff	Level	Level	Diff	Level	Diff
1	8.68					0.297				
2	8.36	10.62	2.26**	10.42	2.06**	0.297	0.358	0.061**	0.370	0.072***
3	7.29	11.97	4.68***	9.17	1.89**	0.269	0.405	0.136***	0.338	0.070**
4	7.04	12.18	5.14***	9.77	2.73***	0.266	0.408	0.142***	0.369	0.103***
5	6.48	12.18	5.70***	9.03	2.55**	0.252	0.406	0.154**	0.352	0.100***
6	5.45	12.34	6.89***	8.30	2.85***	0.216	0.417	0.200***	0.333	0.116***
7	4.64	12.20	7.56***	7.63	2.99***	0.188	0.414	0.226***	0.313	0.125***
8	3.79	12.09	8.30***	7.68	3.89***	0.159	0.421	0.262***	0.326	0.167***
9	3.14	12.53	9.38***	6.96	3.82***	0.136	0.443	0.307***	0.306	0.169***
10	2.65	11.81	9.16***	6.08	3.43***	0.118	0.427	0.309***	0.274	0.155***
11	2.20	10.40	8.19***	5.09	2.88**	0.101	0.391	0.290**	0.237	0.136***

Table 8. Long-Only Momentum

This table reports the net returns and net Sharpe ratios for the long leg of the 12-2, blended, and filtered momentum strategies. It also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 and filter 5 momentum strategies rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Net Return (%)					Net Sharpe				
	12-2	Filtered		Blended		12-2	Filtered		Blended	
	Level	Level	Diff	Level	Diff	Level	Level	Diff	Level	Diff
1	13.64					0.554				
2	13.81	15.03	1.22***	14.22	0.41*	0.561	0.594	0.032***	0.576	0.015
3	13.20	14.93	1.72***	13.89	0.69***	0.536	0.575	0.039**	0.562	0.027**
4	13.21	15.72	2.51***	13.99	0.79***	0.533	0.592	0.059***	0.561	0.028**
5	13.09	16.12	3.03***	13.90	0.81***	0.527	0.600	0.073***	0.555	0.028**
6	12.77	16.36	3.60***	13.87	1.11***	0.515	0.601	0.086***	0.553	0.038***
7	12.54	16.49	3.95***	13.74	1.20***	0.506	0.597	0.092***	0.548	0.042***
8	12.22	16.82	4.60***	13.71	1.49***	0.495	0.608	0.113***	0.547	0.052***
9	11.82	16.81	4.99***	13.24	1.42***	0.479	0.601	0.122***	0.531	0.052***
10	11.44	16.35	4.91***	12.93	1.49***	0.465	0.582	0.118***	0.519	0.055***
11	11.07	14.54	3.47***	12.52	1.45***	0.450	0.521	0.071**	0.504	0.053***

Table 9. Crash Risk

This table reports the skewness, kurtosis, median drawdown size, and median drawdown length for the 12-2, filtered, and blended momentum strategies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the corresponding momentum strategy rebalanced every 5 months.

Horizon	Skewness			Kurtosis			Median Drawdown Size			Median Drawdown Length		
	12-2	Filtered	Blended	12-2	Filtered	Blended	12-2	Filtered	Blended	12-2	Filtered	Blended
1	-1.68			14.30			44.97			79.0		
2	-1.55	-1.57	-1.65	12.35	12.51	13.09	40.68	28.13	30.71	52.0	22.0	37.5
3	-1.57	-1.65	-1.54	12.16	13.55	12.38	45.25	26.98	33.76	52.0	21.0	39.0
4	-1.71	-1.78	-1.66	14.03	15.27	14.14	43.81	29.23	29.68	53.5	22.0	35.5
5	-1.77	-1.69	-1.76	14.33	14.81	15.01	42.11	24.92	32.14	52.5	22.0	37.5
6	-1.79	-1.78	-1.74	14.66	15.36	14.94	45.25	24.63	29.50	52.5	19.0	30.0
7	-1.91	-1.79	-1.93	15.95	16.27	17.60	52.36	26.00	31.33	54.0	19.0	37.5
8	-2.01	-1.85	-2.00	17.02	16.50	18.59	53.11	22.30	29.45	55.0	16.0	29.5
9	-2.14	-1.79	-2.06	18.66	15.95	19.31	54.34	20.44	28.92	73.5	15.0	37.5
10	-2.14	-1.59	-1.98	18.73	11.52	18.31	54.53	20.50	34.32	79.5	15.0	43.0
11	-2.05	-1.02	-1.95	17.70	6.38	17.99	56.53	17.50	37.36	91.5	13.0	43.0

Table 10. Concentrated 12-2 and Blended Momentum Portfolios

This table reports the net returns and net Sharpe ratios for concentrated versions of the 12-2 and blended momentum strategies. The strategies form portfolios of only the highest (and lowest) ranked stocks so that they contain the same number of stocks as in the filtered approach each month at each rebalancing frequency. It also includes a comparison with filtered momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the 12-2 and blended 5 momentum strategies rebalanced every 5 months, and compares it to the filter 5 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Net Return (%)					Net Sharpe				
	Filtered		12-2		Blended		Filtered		12-2	
	Level	Level	Diff	Level	Diff	Level	Level	Diff	Level	Diff
2	13.29	13.05	-0.25	13.40	0.10	0.497	0.488	-0.009	0.505	0.008
3	13.77	12.74	-1.03	14.83	1.06**	0.510	0.477	-0.033	0.551	0.041**
4	14.25	13.96	-0.29	15.01	0.76	0.519	0.520	0.001	0.553	0.033*
5	14.63	13.68	-0.96	15.42	0.79	0.535	0.509	-0.026	0.559	0.024
6	15.04	13.31	-1.73*	15.46	0.42	0.556	0.491	-0.064	0.559	0.003
7	15.07	12.58	-2.49**	15.58	0.51	0.561	0.461	-0.101**	0.559	-0.003
8	16.02	12.11	-3.91***	15.49	-0.53	0.602	0.429	-0.173***	0.553	-0.048
9	15.93	11.23	-4.70***	15.25	-0.68	0.616	0.391	-0.225***	0.541	-0.076*
10	15.10	9.74	-5.37***	13.68	-1.42	0.595	0.325	-0.270***	0.480	-0.115**
11	13.76	7.76	-5.99***	10.86	-2.90*	0.590	0.245	-0.344***	0.370	-0.220***

Table 11. International Evidence

This table reports the net returns and net Sharpe ratios of the 12-2, blended, and filtered momentum strategies for an international sample of stocks from 46 countries during the 1988-2024 period. We winsorize returns at 0.1% and 99.9% each month to account for potential data errors. The table also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 and filter 5 momentum strategies rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Net Return (%)					Net Sharpe				
	12-2	Filtered		Blended		12-2	Filtered		Blended	
	Level	Level	Diff	Level	Diff	Level	Level	Diff	Level	Diff
1	8.34					0.336				
2	8.76	10.45	1.69***	9.62	0.87*	0.374	0.421	0.047*	0.409	0.035
3	7.38	9.37	1.99**	9.61	2.22***	0.323	0.376	0.052	0.420	0.097***
4	6.74	10.28	3.55***	9.66	2.93***	0.303	0.413	0.110**	0.433	0.130***
5	6.08	10.27	4.19***	9.09	3.01***	0.279	0.413	0.135**	0.416	0.137***
6	5.18	9.99	4.82***	8.16	2.99***	0.242	0.407	0.165**	0.383	0.141***
7	4.35	8.90	4.55***	7.48	3.12***	0.208	0.365	0.156*	0.363	0.155***
8	3.87	9.14	5.27***	6.56	2.69***	0.191	0.381	0.190**	0.333	0.142**
9	3.03	8.52	5.48***	5.86	2.83***	0.153	0.367	0.213**	0.310	0.157***
10	2.09	7.67	5.58***	5.14	3.05***	0.108	0.339	0.232**	0.281	0.173***
11	1.27	6.30	5.04**	4.04	2.77***	0.067	0.288	0.221**	0.227	0.160***

Table 12. Machine Learning

This table reports the net returns and net Sharpe ratios for a machine learning investment strategy in the post-publication period. It also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the machine learning strategy rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Net Return (%)			Net Sharpe		
	12-2 Level	Machine Learning Level	Diff	12-2 Level	Machine Learning Level	Diff
1	8.68	7.80	-0.88	0.297	0.383	0.087
2	8.36	10.55	2.20	0.297	0.474	0.177
3	7.29	9.80	2.51	0.269	0.417	0.148
4	7.04	8.76	1.72	0.266	0.353	0.086
5	6.48	8.48	2.00	0.252	0.331	0.079
6	5.45	7.76	2.32	0.216	0.294	0.078
7	4.64	6.67	2.03	0.188	0.250	0.063
8	3.79	6.34	2.55	0.159	0.236	0.077
9	3.14	5.55	2.41	0.136	0.207	0.071
10	2.65	5.40	2.75	0.118	0.200	0.082
11	2.20	5.15	2.95	0.101	0.190	0.089

INTERNET APPENDIX

Momentum at Long Holding Periods

Internet Appendix Table 1. Momentum Using Top and Bottom Three Deciles

This table reports the net returns and net Sharpe ratios of a version of the 12-2, blended, and filtered momentum strategies that goes long stocks in the top three momentum deciles and shorts stocks in the bottom three momentum deciles as in Carhart (1997). The table also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 and filter 5 momentum strategies rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

Horizon	Net Return (%)					Net Sharpe				
	12-2	Filtered		Blended		12-2	Filtered		Blended	
	Level	Level	Diff	Level	Diff	Level	Level	Diff	Level	Diff
1	4.91					0.285				
2	5.55	6.28	0.72***	5.66	0.11	0.331	0.352	0.022	0.337	0.006
3	5.23	6.83	1.60***	5.52	0.29	0.319	0.378	0.060***	0.336	0.017
4	5.12	6.87	1.75***	5.52	0.40	0.317	0.372	0.055**	0.340	0.022
5	4.72	7.21	2.49***	5.31	0.59	0.298	0.387	0.090***	0.334	0.036
6	4.40	7.38	2.98***	5.08	0.67*	0.282	0.396	0.114***	0.327	0.045
7	4.07	7.51	3.44***	4.91	0.84**	0.266	0.404	0.138***	0.326	0.060**
8	3.70	7.71	4.01***	4.71	1.02**	0.247	0.423	0.176***	0.325	0.077**
9	3.23	8.09	4.85***	4.44	1.21***	0.222	0.451	0.229***	0.318	0.096***
10	2.81	8.26	5.44***	4.15	1.33***	0.198	0.468	0.271***	0.307	0.110***
11	2.27	8.33	6.07***	3.79	1.52***	0.162	0.486	0.324***	0.291	0.129***

Internet Appendix Table 2. NYSE 30% Cutoff

This table reports the net returns and net Sharpe ratios of a version of the 12-2, blended, and filtered momentum strategies that excludes stocks below the NYSE 30th percentile market capitalization cutoff which are small and may not be practical for investors to trade. The table also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 and filter 5 momentum strategies rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

	Net Return (%)					Net Sharpe				
Horizon	12-2	Filtered		Blended		12-2	Filtered		Blended	
	Level	Level	Diff	Level	Diff	Level	Level	Diff	Level	Diff
1	8.60					0.327				
2	9.51	10.68	1.18***	9.71	0.20	0.370	0.397	0.027	0.376	0.006
3	8.52	10.94	2.42***	9.33	0.81**	0.341	0.400	0.059***	0.370	0.029*
4	8.30	11.75	3.45***	9.29	0.98**	0.340	0.426	0.086***	0.378	0.038*
5	8.00	12.36	4.36***	9.25	1.25**	0.335	0.448	0.113***	0.386	0.052**
6	7.31	12.52	5.21***	8.95	1.64***	0.315	0.451	0.136***	0.385	0.070***
7	6.81	12.57	5.76***	8.51	1.69***	0.300	0.459	0.159***	0.375	0.076***
8	6.20	13.24	7.04***	8.27	2.08***	0.281	0.486	0.205***	0.378	0.097***
9	5.48	13.49	8.01***	7.56	2.08***	0.254	0.500	0.247***	0.357	0.103***
10	4.75	13.19	8.44***	7.10	2.35***	0.225	0.491	0.266***	0.348	0.122***
11	4.04	11.64	7.59***	6.33	2.29***	0.197	0.491	0.294***	0.324	0.127***

Internet Appendix Table 3. Momentum Including Financials and Utilities

This table reports the net returns and net Sharpe ratios of a version of the 12-2, blended, and filtered momentum strategies that includes financials and utilities, which are excluded from our main analysis. The table also includes a comparison with 12-2 momentum (Diff columns) at corresponding rebalancing frequencies. The horizon column indicates the momentum strategy rebalancing frequency. For example, a horizon of 5 shows results for the blended 5 and filter 5 momentum strategies rebalanced every 5 months, and compares it to 12-2 momentum strategy rebalanced every 5 months. Trading costs are computed by multiplying the annualized roundtrip portfolio turnover by 0.25%. */**/** denotes statistical significance (reported only for the differences in returns and Sharpe ratios) at the 10, 5, and 1 percent levels.

	Net Return (%)					Net Sharpe				
Horizon	12-2	Filtered		Blended		12-2	Filtered		Blended	
	Level	Level	Diff	Level	Diff	Level	Level	Diff	Level	Diff
1	10.74					0.416				
2	11.66	13.18	1.52***	11.53	-0.13	0.472	0.507	0.035**	0.460	-0.012
3	11.01	13.64	2.63***	11.59	0.58	0.460	0.521	0.062***	0.479	0.020
4	10.81	13.95	3.14***	11.57	0.76	0.460	0.526	0.066***	0.487	0.027
5	10.50	14.62	4.13***	11.52	1.03**	0.456	0.553	0.097***	0.497	0.042*
6	9.71	15.03	5.32***	11.33	1.62***	0.432	0.572	0.140***	0.503	0.071***
7	8.91	14.74	5.83***	10.52	1.61***	0.405	0.567	0.162***	0.480	0.076***
8	8.20	15.81	7.61***	10.30	2.11***	0.383	0.615	0.232***	0.485	0.102***
9	7.27	15.88	8.61***	9.65	2.38***	0.348	0.638	0.289***	0.470	0.122***
10	6.46	15.33	8.86***	8.99	2.52***	0.317	0.615	0.298***	0.454	0.137***
11	5.65	14.86	9.21***	8.19	2.54***	0.285	0.671	0.386***	0.432	0.147***